Theories as Regularizers*

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Abstract

Machine-learning methods for return prediction ignore economic theory. Structural asset-pricing models encode deep knowledge about expected returns but are individually misspecified. We let the data decide which theories help and how much. We embed over fifty model-implied restrictions—from consumption-based, production-based, intermediary, behavioral, and demand-system frameworks—as soft penalties in a kernel ridge regression estimator. Each restriction carries its own penalty weight, selected by cross-validation: theories that hurt prediction are silenced; theories that help are amplified. The result is the first empirical ranking of economic theories by their marginal out-of-sample forecasting value. Theory-informed penalization delivers higher out-of-sample predictive accuracy and Sharpe ratios than both agnostic machine learning and standard kernel methods. Consumption-based and intermediary restrictions contribute the most. Theory helps disproportionately when data are scarce and during crises—precisely when investors need guidance most.

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1 Introduction

Two traditions compete for the task of predicting asset returns. Statistical machine-learning methods—neural networks, random forests, gradient-boosted trees—estimate flexible functions of observable characteristics and deliver strong out-of-sample (OOS) performance, yet they treat economic theory as irrelevant input (Gu et al., 2020). Structural asset-pricing models—CAPM, long-run risk, habit formation, intermediary-based pricing—derive sharp restrictions on the pricing kernel and expected returns, yet each model is misspecified in isolation. One tradition has power without guidance; the other has guidance without robustness. Neither alone is satisfactory.

We propose a framework that draws on both. Our estimator, Theory-Informed Kernel Ridge Regression (Theory-KRR), embeds fifty-six model-implied structural restrictions as soft penalties in the KRR objective. Each restriction j enters with its own penalty multiplier μ_j, and the entire vector $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_J)$ is selected by cross-validation. The mechanism is automatic: restrictions from misspecified theories receive $\mu_j \to 0$—the data shut them down—while restrictions from empirically useful theories receive $\mu_j > 0$, steering the estimator toward economically sensible regions of the reproducing kernel Hilbert space $\mathcal{H}$. No single structural model needs to be correct. The data adjudicate.

The restrictions we impose are not generic shape constraints such as monotonicity or convexity (Bryzgalova et al., 2023b). They are genuine model-implied restrictions: Euler-equation moment conditions from consumption-based and intermediary SDFs, production first-order conditions, and demand-system equilibrium conditions, each derived from a specific economic theory. By cataloging fifty-six such restrictions across consumption-based, production-based, intermediary, behavioral, and demand-system frameworks, we construct a comprehensive menu of theoretical guidance. Cross-validation then selects which theories—and which specific restrictions within each theory—contribute to OOS prediction.

A conceptual parallel clarifies the logic. The LLM-Lasso of Bybee et al. (2025) replaces agnostic ℓ_1 penalties with heterogeneous, text-informed penalties, letting cross-validation

arbitrate between prior and data. Our approach applies the same principle—informed, heterogeneous penalization with data-driven selection—but operates in function space with economic-theory priors rather than in parameter space with text priors.

What does this framework deliver? Theory-KRR outperforms standard KRR and several benchmark ML methods in OOS return prediction, measured by R^2_{OOS} and portfolio Sharpe ratios. The improvement is not uniform: theory-informed penalization helps most in small samples and during financial crises, precisely when purely statistical methods overfit and economic structure is most valuable. Beyond predictive gains, the cross-validated multipliers $\boldsymbol{\mu}$ produce an empirical ranking of economic theories by their marginal OOS forecasting value. Consumption-based restrictions (Campbell and Cochrane, 1999; Bansal and Yaron, 2004) and intermediary-based restrictions (He and Krishnamurthy, 2013; Adrian et al., 2014) receive the largest multipliers, indicating that these theories provide the most useful guidance for return prediction. Production-based and behavioral restrictions contribute modestly. Several models receive μ_j indistinguishable from zero, consistent with those theories being empirically irrelevant for forecasting in our sample.

This paper makes three contributions. First, we develop a unified framework that embeds fifty-six structural restrictions from diverse asset-pricing models as soft penalties in a single KRR estimator. The framework nests standard KRR (all $\mu_j = 0$) and hard-constrained estimation (any $\mu_j \to \infty$) as special cases, and it accommodates restrictions of heterogeneous functional form—Euler equation moment conditions, production first-order conditions, and demand-system equilibrium conditions—within a single objective. Second, we propose joint estimation of the nonparametric function and the structural parameters that enter the penalties, allowing theory and data to interact rather than treating structural parameters as fixed inputs. Third, we produce the first empirical ranking of economic theories by their marginal contribution to OOS forecasting, providing a new lens through which to evaluate the empirical content of competing asset-pricing models.

Our work speaks to several strands of the literature. The ML asset-pricing literature,

initiated by Gu et al. (2020), demonstrates that flexible nonparametric methods substantially outperform linear models in predicting the cross-section of returns. Subsequent work explores neural-network architectures (Chen et al., 2024), autoencoder factor models with no-arbitrage constraints (Gu et al., 2021), conditional factor models (Kelly et al., 2019), and Gaussian process regression applied to equity returns (Filipović and Pasricha, 2022). Kelly et al. (2024) provide a theoretical foundation for this success, showing that complex, overparameterized models are not penalized by benign overfitting and systematically outperform simpler ones. These methods are powerful but agnostic about economic structure. Our contribution is to show that injecting economic theory into the regularization of a kernel method yields both predictive gains and economic interpretability.

A growing body of work recognizes that theory and statistics are complements, not substitutes, for return prediction. The paper closest to ours in spirit is Chen et al. (2026), who show that pre-training neural networks on synthetic data generated by structural models—and then fine-tuning on empirical data—substantially improves option-pricing accuracy, even when the structural model is misspecified. Our approach shares the same fundamental insight—misspecified theory still helps—but differs in three important respects. First, we use soft penalization in the objective function rather than pre-training: structural content enters directly as penalty terms with cross-validated weights, not as an initialization from synthetic data. Second, we target the cross-section of equity returns rather than option pricing. Third, the convexity of our KRR objective guarantees a global optimum and yields tractable asymptotics, whereas neural-network pre-training is inherently non-convex. Most importantly, our penalty weights provide a continuous measure of each theory's marginal contribution to prediction, whereas transfer learning offers a binary use-or-not decision for each structural model.

A second closely related paper is Bryzgalova et al. (2023a), who develop a Bayesian framework that averages over 2.25 quadrillion linear asset-pricing models formed by subsets of 51 observable factors. Their identification-strength-based priors impose heterogeneous

shrinkage—factors that are weakly identified receive stronger shrinkage—and the resulting Bayesian model-averaging stochastic discount factor dominates individual factor models out of sample. Like us, they produce an empirical ranking of economic models. The key differences are the objects being ranked and the estimation framework. Bryzgalova, Huang, and Julliard rank observable factor portfolios within a linear SDF; we rank structural restrictions from named theoretical models within a nonparametric RKHS. Their priors are calibrated from factor identification strength; our penalty weights are selected by temporal cross-validation, tying the ranking directly to out-of-sample predictive performance.

More broadly, the question of which factors or models survive in a high-dimensional setting has been explored from several angles. Kozak et al. (2020) propose economically motivated shrinkage of SDF coefficients proportional to the eigenvalues of characteristic-based principal components, motivated by no-arbitrage bounds on Sharpe ratios. Feng et al. (2020) develop a double-selection LASSO procedure to test whether newly proposed factors have marginal explanatory power beyond existing ones. Harvey et al. (2016) document severe data snooping in factor discovery and argue for higher significance thresholds. Barillas and Shanken (2018) and Kan et al. (2013) provide formal Bayesian and frequentist frameworks for pairwise model comparison. Our paper complements this literature by moving from factor-level selection in linear models to theory-level selection in a nonparametric estimator, where the "theories" are entire families of structural restrictions rather than individual tradable factors.

The econometrics literature on constrained nonparametric estimation provides our mathematical foundation. Babii et al. (2024) develop the theory of penalized kernel regression with functional constraints, establishing convergence rates and conditions under which constraints improve estimation. Gabaix et al. (2024) show how demand-system equilibrium conditions can be imposed on asset-pricing estimators, and Bryzgalova et al. (2023b) impose shape restrictions (monotonicity, convexity) on pricing-kernel estimators. In the penalized GMM literature, Caner (2009) and Caner and Zhang (2014) develop LASSO-type and elastic-net

GMM estimators with oracle properties for moment selection. Freyberger et al. (2020) use adaptive group LASSO to select characteristics nonparametrically. Our work extends this line by moving from reduced-form shape restrictions to model-implied structural restrictions, and by allowing the data to select among competing restrictions via cross-validation rather than imposing all restrictions with equal weight. The representer theorem, generalized to our multi-penalty setting following Schölkopf and Smola (2002) and the foundational results of Kimeldorf and Wahba (1971) and Wahba (1990), ensures that the solution lies in the span of kernel evaluations even with the added structural penalties.

Our choice of kernel ridge regression as the base estimator is grounded in recent advances in kernel learning theory. Under a source condition $f^* \in [\mathcal{H}]^s$ with $0 < s \leq 1$, KRR achieves the minimax-optimal convergence rate (Zhang et al., 2025). For smoother targets ($s > 1$), KRR exhibits a saturation effect—it cannot exploit additional regularity regardless of how the regularization parameter is tuned (Li and Zhang, 2023). This saturation is a fundamental limitation of all spectral methods, which lack the feature-learning capability of neural networks (Barzilai et al., 2024). Our structural penalties address this limitation by effectively reducing the function class to a smaller set consistent with economic theory, mimicking feature alignment within the kernel framework. The sharp asymptotics of Hu and Lu (2022) further reveal that KRR exhibits multiple descent phenomena in the polynomial scaling regime $n \approx d^k$, and the geometrical analysis of Zhang et al. (2024a) provides non-asymptotic excess risk bounds that govern the bias-variance trade-off in our setting. Importantly, Wang and Jing (2022) establish that Gaussian process regression—equivalent to KRR under the appropriate correspondence—achieves optimal rates even under misspecified kernel smoothness, provided the imposed smoothness exceeds the true smoothness, lending robustness to our kernel choice.

The inferential foundations for testing whether individual structural penalties contribute to prediction rest on the asymptotic theory of Fischer and Steinwart (2024), who establish asymptotic normality for linear functionals of KRR estimators, including point evaluations

and integrated functionals. Their results yield a parametric convergence rate n^{-1} for sufficiently smooth functionals, enabling formal tests of $H_0 : \mu_j = 0$. For uniform inference, Singh (2024) constructs valid confidence bands for kernel regression estimators using a Gaussian multiplier bootstrap, providing the distributional theory needed for simultaneous inference across evaluation points. The consistency of generalized cross-validation for selecting λ_{stat} in the polynomial scaling regime, recently established by Zhang et al. (2024b), provides theoretical justification for our temporal cross-validation strategy. Together, these results place the statistical properties of Theory-KRR—convergence rates, inference on μ_j, and cross-validation consistency—on rigorous foundations.

The paper proceeds as follows. Section 2 presents the mathematical framework, defining the Theory-KRR objective and establishing its properties. Section 3 catalogs the fifty-six structural restrictions we draw from the asset-pricing literature, organized by model class. Section 4 addresses practical estimation, including cross-validation of the multiplier vector $\boldsymbol{\mu}$, computational strategies, and implementation details. Section 5 describes the data and baseline models. Section 6 presents the main empirical results: OOS predictive performance, the theory ranking, and robustness exercises. Section 7 concludes.

2 The Theory-Informed Kernel Ridge Regression Estimator

This section develops the econometric framework. Section 2.1 introduces notation and the prediction environment. Section 2.2 defines the theory-informed KRR objective. Sections 2.3–2.5 formalize the three structural penalty types and establish the representer theorem and finite-dimensional reduction that make the estimator computationally tractable. Section 2.6 shows that the framework nests standard KRR and constrained GMM as special cases. Sections 2.7–2.9 address implementation, misspecification robustness, and inference.

2.1 Setup and Notation

We observe a panel of N assets over T periods. Let $R_{i,t+1}$ denote the excess return of asset i realized at date $t+1$, and let $\mathbf{X}_{i,t} \in \mathbb{R}^p$ collect firm-level characteristics and macroeconomic state variables observable at date t. The total sample size is $n = NT$.

Assumption 1 (Data-generating process)**.** There exists a measurable function $f^* : \mathbb{R}^p \to \mathbb{R}$ such that

$$R_{i,t+1} = f^*(\mathbf{X}_{i,t}) + \varepsilon_{i,t+1}, \qquad \mathbb{E}[\varepsilon_{i,t+1} \mid \mathbf{X}_{i,t}] = 0, \tag{1}$$

where $f^*(\mathbf{X}_{i,t}) = \mathbb{E}[R_{i,t+1} \mid \mathbf{X}_{i,t}]$ is the conditional expected excess return.

We approximate f^* within a reproducing kernel Hilbert space (RKHS).

Definition 2 (RKHS)**.** Let $k : \mathbb{R}^p \times \mathbb{R}^p \to \mathbb{R}$ be a symmetric, positive-definite kernel. The RKHS $\mathcal{H}$ associated with k is the completion of $\operatorname{span}\{k(\mathbf{x}, \cdot) : \mathbf{x} \in \mathbb{R}^p\}$ under the inner product satisfying the reproducing property $f(\mathbf{x}) = \langle f, k(\mathbf{x}, \cdot)\rangle_{\mathcal{H}}$ for all $f \in \mathcal{H}$. The RKHS norm $\|f\|_{\mathcal{H}}^2 = \langle f, f\rangle_{\mathcal{H}}$ measures the complexity of f.

We stack observations using the index $s = (i, t)$, so that $\mathbf{X}_s \equiv \mathbf{X}_{i,t}$ and $R_s \equiv R_{i,t+1}$ for $s = 1, \ldots, n$. The $n \times n$ kernel (Gram) matrix is $\mathbf{K}$, with $[\mathbf{K}]_{s,s'} = k(\mathbf{X}_s, \mathbf{X}_{s'})$. The return vector is $\mathbf{R} = (R_1, \ldots, R_n)^\top \in \mathbb{R}^n$.

2.2 The Estimator

Our estimator augments the standard KRR objective with a collection of structural penalty terms, each encoding restrictions implied by a specific economic model.

Definition 3 (Theory-Informed KRR)**.** Let $\{C_j\}_{j=1}^J$ be a collection of structural penalty functionals $C_j : \mathcal{H} \to \mathbb{R}_+$, where each $C_j(f) \geq 0$ with equality when f satisfies the restrictions of economic model j. Let $\lambda_{\text{stat}} > 0$ and $\mu_j \geq 0$, $j = 1, \ldots, J$. The *theory-informed KRR*

estimator is

$$\hat{f} \;=\; \arg\min_{f\in\mathcal{H}} \underbrace{\frac{1}{n}\sum_{s=1}^{n}\big(R_s - f(\mathbf{X}_s)\big)^2}_{\text{prediction loss}} + \underbrace{\lambda_{\text{stat}}\,\|f\|^2_{\mathcal{H}}}_{\text{statistical regularization}} + \underbrace{\sum_{j=1}^{J}\mu_j\, C_j(f)}_{\text{structural penalties}} \;. \tag{2}$$

The first two terms constitute standard kernel ridge regression: the squared-error prediction loss trades off against the RKHS-norm penalty, which controls the smoothness of f. The third term is new. Each C_j penalizes deviations from a specific economic model's predictions, and the associated multiplier μ_j governs how strongly that model's restrictions influence the estimate. When $\mu_j = 0$, model j is ignored; as $\mu_j \to \infty$, the restrictions of model j are imposed exactly.

Remark 4 (Interpretation of multipliers). The vector $(\lambda_{\text{stat}}, \mu_1, \ldots, \mu_J)$ balances agnostic statistical regularization against theory-informed regularization. A large λ_{stat} relative to μ_j yields an estimate close to standard KRR; large μ_j relative to λ_{stat} forces the estimator to respect model j's restrictions even if doing so worsens in-sample fit. Cross-validation selects the balance that maximizes out-of-sample predictive accuracy (Section 2.7).

2.3 Structural Penalty Types

We formalize three classes of structural penalties. Each encodes restrictions from a distinct family of asset-pricing models. A key requirement—verified explicitly below—is that every $C_j(f)$ depends on f only through its evaluations at the observed data points $\{f(\mathbf{X}_s)\}_{s=1}^n$.

2.3.1 Type A: Euler Equation Penalties

Let model j specify a stochastic discount factor (SDF) $M_{j,t+1} = M_j(\theta_j, \mathbf{X}_t, \mathbf{X}_{t+1})$ parameterized by θ_j, with a preliminary estimator $\hat{\theta}_j$ obtained by GMM or maximum likelihood. The Euler equation requires $\mathbb{E}_t[M_{j,t+1}\, R^{\text{gross}}_{i,t+1}] = 1$ for every asset i and date t, where $R^{\text{gross}}_{i,t+1} = f(\mathbf{X}_{i,t}) + R^f_t$ is the gross return and R^f_t is the risk-free rate.

Definition 5 (Euler equation penalty)**.** For SDF model j with pre-estimated parameter $\hat{\theta}_j$, define

$$C_j^{\text{Euler}}(f) \;=\; \sum_{t=1}^{T} \left\| \frac{1}{N} \sum_{i=1}^{N} M_j\big(\hat{\theta}_j, \mathbf{X}_t, \mathbf{X}_{t+1}\big) \cdot \big(f(\mathbf{X}_{i,t}) + R_t^f\big) - 1 \right\|^2 . \tag{3}$$

The penalty is zero precisely when the cross-sectional Euler equation holds at every date t given $\hat{\theta}_j$. Since $M_j(\hat{\theta}_j, \mathbf{X}_t, \mathbf{X}_{t+1})$ and R_t^f are data, the penalty depends on f only through $\{f(\mathbf{X}_{i,t})\}_{i,t}$.

Remark 6 (Coverage). Type A penalties accommodate any model that specifies an SDF $M_{j,t+1}$. This includes consumption-based models (CCAPM, long-run risk, rare disasters, habit formation), intermediary-based models where the SDF is linear in a factor $F_{j,t}$ (i.e., $M_{j,t+1} = a_j + b_j F_{j,t+1}$, as in He and Krishnamurthy, 2013 and Adrian et al., 2014), information and learning models, and behavioral models that specify a modified pricing kernel. A linear factor model $M_{t+1} = 1 - \mathbf{b}^\top(\mathbf{F}_{t+1} - \mathbb{E}[\mathbf{F}])$ is a linear SDF, so any factor-based restriction reduces to a Type A Euler equation penalty without requiring pre-estimated betas—only the factor time series $F_{j,t}$ is needed. Each distinct SDF specification generates a separate penalty C_j^{Euler}.

2.3.2 Type B: Production and Investment FOC Penalties

Production-based asset pricing relates expected returns to observable investment and profitability variables through firms' first-order conditions. Let model j imply a mapping $g_j : \mathbb{R}^{d_j} \to \mathbb{R}$ from investment characteristics $\mathbf{Z}_{i,t}^{(j)}$ (e.g., investment-to-capital ratio $I_{i,t}/K_{i,t}$, return on equity $\text{ROE}_{i,t}$, asset growth) to expected returns.

Definition 7 (Production FOC penalty)**.** For production model j with predicted returns $g_j(\mathbf{Z}_{i,t}^{(j)})$, define

$$C_j^{\text{prod}}(f) \;=\; \frac{1}{n} \sum_{s=1}^{n} \big(f(\mathbf{X}_s) - g_j\big(\mathbf{Z}_s^{(j)}\big)\big)^2. \tag{4}$$

The penalty is a mean-squared deviation between the nonparametric estimate and the production model's implied expected return. Since $g_j(\mathbf{Z}_s^{(j)})$ is computed from data, the penalty

depends on f only through $\{f(\mathbf{X}_s)\}_{s=1}^n$.

Remark 8 (Nesting the q-theory framework). When g_j encodes the first-order condition from the investment CAPM or the q-factor model, Type B penalties embed the production approach to asset pricing directly into the nonparametric objective.

2.3.3 Type D: Demand-System Equilibrium Penalties

Demand-system approaches to asset pricing derive equilibrium prices from investors' portfolio demand functions. Let $D_i : \mathbb{R}^N \to \mathbb{R}^N$ map the vector of expected returns to investor i's optimal portfolio weights, and let $w_i > 0$ denote investor i's wealth share, with the index now running over investors rather than assets. Let $\mathbf{S}_t \in \mathbb{R}^N$ denote the vector of asset market-capitalization shares (the supply side) at date t.

Definition 9 (Demand-system equilibrium penalty)**.** For demand-system model j with demand functions $\{D_i^{(j)}\}$ and observed supplies $\{\mathbf{S}_t\}$, let $\hat{\mathbf{f}}_t = (f(\mathbf{X}_{1,t}), \ldots, f(\mathbf{X}_{N,t}))^\top$ collect expected-return evaluations. Define

$$C_j^{\text{demand}}(f) \;=\; \sum_{t=1}^{T} \left\| \sum_i w_i^{(j)} \, D_i^{(j)}\big(\hat{\mathbf{f}}_t\big) - \mathbf{S}_t \right\|^2 . \tag{5}$$

The penalty measures the squared market-clearing failure: it is zero when the demand system clears at the predicted expected returns. It depends on f only through $\{f(\mathbf{X}_{i,t})\}_{i,t}$.

Remark 10 (Connections to the demand-system literature). The demand-system approach models institutional investors' demand as a function of characteristics and expected returns. Type D penalties encode the equilibrium restriction that aggregate demand equals supply, translating the demand-system estimator's logic into a penalty within our nonparametric framework. When $D_i^{(j)}$ is linear in expected returns, C_j^{demand} is quadratic in $\hat{\mathbf{f}}_t$ and the overall problem remains convex.

2.4 The Representer Theorem

The following result establishes that the infinite-dimensional optimization problem (2) admits a finite-dimensional solution, despite the presence of the structural penalties.

Proposition 11 (Representer theorem for theory-informed KRR)**.** *Let $\mathcal{H}$ be an RKHS with kernel k. Suppose each structural penalty $C_j : \mathcal{H} \to \mathbb{R}_+$ depends on f only through the vector of evaluations $(f(\mathbf{X}_1), \ldots, f(\mathbf{X}_n))^\top \in \mathbb{R}^n$. That is, there exist functions $\tilde{C}_j : \mathbb{R}^n \to \mathbb{R}_+$ such that*

$$C_j(f) = \tilde{C}_j\big(f(\mathbf{X}_1), \ldots, f(\mathbf{X}_n)\big) \qquad \text{for all } f \in \mathcal{H},\ j = 1, \ldots, J. \tag{6}$$

Then any minimizer of (2) admits the representation

$$\hat{f}(\cdot) = \sum_{s=1}^{n} \hat{\alpha}_s\, k(\mathbf{X}_s, \cdot), \tag{7}$$

for some $\hat{\boldsymbol{\alpha}} = (\hat{\alpha}_1, \ldots, \hat{\alpha}_n)^\top \in \mathbb{R}^n$.

Proof. The proof is given in Appendix A. □

Remark 12 (Verification for Types A, B, and D). The condition (6) holds for all three penalty types:

(a) *Type A (Euler):* $C_j^{\text{Euler}}(f)$ depends on f through $\{f(\mathbf{X}_{i,t})\}_{i,t}$, since $M_j(\hat{\theta}_j, \mathbf{X}_t, \mathbf{X}_{t+1})$ and R_t^f are data.

(b) *Type B (Production):* $C_j^{\text{prod}}(f)$ is explicitly a function of $\{f(\mathbf{X}_s)\}_s$ and data $\{g_j(\mathbf{Z}_s^{(j)})\}_s$.

(c) *Type D (Demand):* $C_j^{\text{demand}}(f)$ depends on f through $\hat{\mathbf{f}}_t = (f(\mathbf{X}_{1,t}), \ldots, f(\mathbf{X}_{N,t}))^\top$.

In each case, the penalty is a function of the n-dimensional evaluation vector, so Proposition 11 applies. This extends classical representer theorems to accommodate theory-informed structural penalties.

Remark 13 (Evaluation-dependence and variable coverage). A key feature of the penalty structure is that each $C_j(f)$ constrains f only through its evaluations at the observed data points $\{f(\mathbf{X}_s)\}_{s=1}^n$. This has an important implication for the interpretation of penalties that involve only a subset of the characteristic vector $\mathbf{X}$. Consider a production-based penalty $C_j^{\text{prod}}(f)$ that depends on investment-to-capital ratio (I/K) and return on equity (ROE) but not on consumption growth. This penalty does *not* impose a zero restriction on f's dependence on consumption—it is simply silent about consumption. The penalty penalizes deviations of $f(\mathbf{X}_s)$ from the production model's prediction $g_j(\mathbf{Z}_s^{(j)})$ at each observed point, but it leaves f free to depend on any variable in $\mathbf{X}$ to match those targets. More generally, each C_j constrains f along specific dimensions of the characteristic space while leaving other dimensions unrestricted. The combined penalty $\sum_j \mu_j C_j(f)$ achieves broad coverage by assembling restrictions from different models, each constraining different dimensions of $\mathbf{X}$.

2.5 Finite-Dimensional Reduction

By Proposition 11, it suffices to optimize over $\boldsymbol{\alpha} \in \mathbb{R}^n$. Substituting $f(\cdot) = \sum_s \alpha_s\, k(\mathbf{X}_s, \cdot)$ into the objective yields the following.

Proposition 14 (Finite-dimensional problem)**.** *Under the representer theorem (Proposition 11), the theory-informed KRR estimator* (2) *reduces to*

$$\hat{\boldsymbol{\alpha}} \;=\; \arg\min_{\boldsymbol{\alpha}\in\mathbb{R}^n} \; \frac{1}{n}\|\mathbf{R} - \mathbf{K}\boldsymbol{\alpha}\|^2 \;+\; \lambda_{\text{stat}}\, \boldsymbol{\alpha}^\top \mathbf{K}\, \boldsymbol{\alpha} \;+\; \sum_{j=1}^{J} \mu_j\, \tilde{C}_j(\mathbf{K}\boldsymbol{\alpha}), \tag{8}$$

where $\mathbf{K}$ *is the* $n \times n$ *kernel matrix,* $\hat{\mathbf{f}} = \mathbf{K}\boldsymbol{\alpha}$ *is the* $n \times 1$ *vector of fitted values, and* $\tilde{C}_j(\hat{\mathbf{f}}) = C_j(f)$ *expresses each penalty in terms of fitted values.*

Proof. The proof is given in Appendix A. □

2.5.1 Convexity and First-Order Conditions for Quadratic Penalties

When all active structural penalties are quadratic in the fitted values $\hat{\mathbf{f}} = \mathbf{K}\boldsymbol{\alpha}$, the finite-dimensional problem (8) becomes a convex quadratic program with a closed-form solution.

For Type A penalties, define the $T \times n$ matrix $\mathbf{A}_j$ with rows indexed by t:

$$[\mathbf{A}_j]_{t,s} = \begin{cases} \dfrac{1}{N} M_j\big(\hat{\theta}_j, \mathbf{X}_t, \mathbf{X}_{t+1}\big) & \text{if } s = (i,t) \text{ for some } i, \\ 0 & \text{otherwise,} \end{cases} \tag{9}$$

and the $T \times 1$ vector $\mathbf{b}_j$ with $[\mathbf{b}_j]_t = 1 - \frac{1}{N}\sum_i M_j(\hat{\theta}_j, \mathbf{X}_t, \mathbf{X}_{t+1})\, R_t^f$. Then

$$C_j^{\text{Euler}}(f) = \|\mathbf{A}_j\,\hat{\mathbf{f}} - \mathbf{b}_j\|^2 = (\mathbf{A}_j\,\mathbf{K}\boldsymbol{\alpha} - \mathbf{b}_j)^\top(\mathbf{A}_j\,\mathbf{K}\boldsymbol{\alpha} - \mathbf{b}_j). \tag{10}$$

For Type B penalties, let $\mathbf{g}_j \in \mathbb{R}^n$ with $[\mathbf{g}_j]_s = g_j(\mathbf{Z}_s^{(j)})$. Then

$$C_j^{\text{prod}}(f) = \frac{1}{n}\|\mathbf{K}\boldsymbol{\alpha} - \mathbf{g}_j\|^2. \tag{11}$$

Proposition 15 (First-order conditions)**.** *Suppose every active penalty ($\mu_j > 0$) is of Type A or B (or more generally, quadratic in $\hat{\mathbf{f}}$). Define the aggregated penalty matrix*

$$\boldsymbol{\Omega} \;=\; \sum_{j\in\mathcal{J}^A} \mu_j\,\mathbf{A}_j^\top\mathbf{A}_j \;+\; \sum_{j\in\mathcal{J}^B} \frac{\mu_j}{n}\,\mathbf{I}_n, \tag{12}$$

where $\mathcal{J}^A$ and $\mathcal{J}^B$ index Type A and B penalties respectively, and the aggregated target vector

$$\boldsymbol{\omega} \;=\; \sum_{j\in\mathcal{J}^A} \mu_j\,\mathbf{A}_j^\top\mathbf{b}_j \;+\; \sum_{j\in\mathcal{J}^B} \frac{\mu_j}{n}\,\mathbf{g}_j. \tag{13}$$

Then the first-order conditions for (8) are

$$\left[\frac{1}{n}\mathbf{K} + \lambda_{\text{stat}}\,\mathbf{I}_n + \boldsymbol{\Omega}\,\mathbf{K}\right]\boldsymbol{\alpha} \;=\; \frac{1}{n}\mathbf{R} + \boldsymbol{\omega}, \tag{14}$$

and when $\mathbf{K}$ *is strictly positive definite, the unique solution is*

$$\hat{\boldsymbol{\alpha}} \;=\; \left[\frac{1}{n}\,\mathbf{K} + \lambda_{\text{stat}}\,\mathbf{I}_n + \boldsymbol{\Omega}\,\mathbf{K}\right]^{-1} \left(\frac{1}{n}\,\mathbf{R} + \boldsymbol{\omega}\right). \tag{15}$$

Proof. The proof is given in Appendix A. □

Remark 16 (Computational cost). The dominant cost is solving the $n \times n$ linear system (15), which requires $O(n^3)$ operations. When n is large, one can exploit the eigendecomposition $\mathbf{K} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^\top$ (computed once at cost $O(n^3)$) and solve in the rotated basis, where the matrix becomes diagonal plus a low-rank perturbation from the Type A penalties. Alternatively, random Fourier features or Nyström approximations reduce the effective dimension.

Remark 17 (Type D penalties and convexity). When demand functions $D_i^{(j)}$ are linear in expected returns, C_j^{demand} is quadratic in $\hat{\mathbf{f}}$ and the problem remains a convex QP with a structure analogous to (14). For nonlinear demand systems, the problem is generally nonconvex and requires iterative methods (e.g., projected gradient descent or alternating minimization), though convexity of the remaining terms provides a useful warm start.

Remark 18 (Uniqueness under nonconvex penalties). When some C_j are nonconvex in the fitted values—as arises with nonlinear SDF specifications in Type A penalties—the overall objective (8) may have multiple local minima. A sufficient condition for uniqueness is that the statistical regularization dominates the nonconvex curvature: if the spectral norm of the Hessian of $\sum_j \mu_j \tilde{C}_j(\hat{\mathbf{f}})$ at any critical point is bounded by $2\lambda_{\text{stat}}\,\lambda_{\min}(\mathbf{K})/n$, where $\lambda_{\min}(\mathbf{K})$ is the smallest eigenvalue of $\mathbf{K}$, then the overall Hessian remains positive definite and the solution is unique. This condition is verifiable in practice from the kernel matrix and the penalty structure. The representer theorem (Proposition 11) holds regardless of convexity—it requires only that $\lambda_{\text{stat}} > 0$ and that penalties depend on f through evaluations—so the finite-dimensional reduction is valid even when the reduced problem has multiple critical points (Bodin, 2024).

2.6 Nesting Structure

The framework nests two important special cases, clarifying the relationship between theory-informed KRR and existing methods.

Proposition 19 (Nesting)**.** *The theory-informed KRR estimator* (2) *nests the following:*

(i) ***Standard KRR.*** *Setting $\mu_j = 0$ for all $j = 1, \ldots, J$ yields*

$$\hat{f}^{\mathrm{KRR}} = \arg\min_{f \in \mathcal{H}} \frac{1}{n} \sum_s (R_s - f(\mathbf{X}_s))^2 + \lambda_{\mathrm{stat}} \|f\|_{\mathcal{H}}^2,$$

the standard kernel ridge regression estimator.

(ii) ***Constrained minimum-distance / GMM.*** *Setting $\lambda_{\mathrm{stat}} = 0$ and activating a single Type A penalty with $\mu_j \to \infty$ yields*

$$\hat{f}^{\mathrm{GMM}} = \arg\min_{f \in \mathcal{H}} \frac{1}{n} \sum_s (R_s - f(\mathbf{X}_s))^2 \quad s.t. \quad C_j^{\mathrm{Euler}}(f) = 0,$$

which is a nonparametric generalization of GMM-based SDF estimation. When $\mathcal{H}$ is restricted to parametric functions, this recovers classical GMM.

Proof. The proof is given in Appendix A. □

Remark 20 (Weak dominance). Since standard KRR corresponds to $\mu_j = 0$ for all j, and the cross-validated theory-informed KRR is free to select $\hat{\mu}_j = 0$ if theory j is unhelpful, the theory-informed estimator weakly dominates standard KRR in out-of-sample prediction loss—up to the granularity of the hyperparameter search.

2.7 Cross-Validation for Hyperparameter Selection

The hyperparameter vector $\boldsymbol{\lambda} = (\lambda_{\mathrm{stat}}, \mu_1, \ldots, \mu_J)$ is $(J+1)$-dimensional. We introduce grouped penalties and temporal cross-validation to make the selection tractable.

We assign each model j to one of G groups based on its economic type. Let $g : \{1, \ldots, J\} \to \{1, \ldots, G\}$ denote the group assignment, and impose $\mu_j = \mu_{g(j)}$ for all j. In our empirical application, $G = 8$, corresponding to the three penalty types (A, B, D) each potentially with multiple variants (e.g., consumption-based vs. intermediary-based SDFs within Type A). This reduces the hyperparameter space from $(J + 1)$ to $(G + 1) = 9$ dimensions.

Financial panel data exhibit strong serial dependence, so we use leave-one-year-out (LOYO) cross-validation to select $\boldsymbol{\lambda}$. Partition the sample into T_{yr} calendar years $\mathcal{T}_1, \ldots, \mathcal{T}_{T_{\text{yr}}}$. For fold ℓ, let $\mathcal{S}_{-\ell} = \{(i, t) : t \notin \mathcal{T}_\ell\}$ be the training set and $\mathcal{S}_\ell = \{(i, t) : t \in \mathcal{T}_\ell\}$ the validation set. The cross-validated objective is

$$\text{CV}(\boldsymbol{\lambda}) \;=\; \frac{1}{T_{\text{yr}}} \sum_{\ell=1}^{T_{\text{yr}}} \frac{1}{|\mathcal{S}_\ell|} \sum_{s \in \mathcal{S}_\ell} \big(R_s - \hat{f}_{-\ell}(\mathbf{X}_s; \boldsymbol{\lambda})\big)^2, \tag{16}$$

where $\hat{f}_{-\ell}(\cdot; \boldsymbol{\lambda})$ is the theory-informed KRR estimator trained on $\mathcal{S}_{-\ell}$ with hyperparameters $\boldsymbol{\lambda}$.

We minimize $\text{CV}(\boldsymbol{\lambda})$ by random search over a 9-dimensional log-uniform grid: $\log_{10} \lambda_{\text{stat}} \in [-6, 2]$ and $\log_{10} \mu_g \in [-6, 2]$ for $g = 1, \ldots, G$. We draw $B = 500$ random configurations and evaluate each by solving the finite-dimensional problem (8) T_{yr} times (once per fold). The selected hyperparameters are $\hat{\boldsymbol{\lambda}} = \arg\min_{\boldsymbol{\lambda}} \text{CV}(\boldsymbol{\lambda})$.

Remark 21 (Computational considerations). Each fold requires solving a linear system of size $|\mathcal{S}_{-\ell}|$, which is $O(((T_{\text{yr}} - 1) \cdot N/T_{\text{yr}})^3)$ per fold. With $T_{\text{yr}} \approx 30$–60 years and $N \approx 3{,}000$–10,000 assets, this is demanding but feasible with precomputed kernel eigendecompositions and parallelization across folds and random draws. The total cost scales as $O(B \cdot T_{\text{yr}} \cdot n_{\text{fold}}^3)$ where $n_{\text{fold}} = |\mathcal{S}_{-\ell}|$.

2.8 Misspecification Robustness

A central concern with theory-informed estimation is that imposing restrictions from a misspecified model may worsen predictions. The following result shows that cross-validation

provides an automatic safeguard.

Proposition 22 (Misspecification robustness)**.** *Suppose model j is misspecified in the sense that the true conditional expected return function f^* satisfies $C_j(f^*) > 0$. Under the following regularity conditions:*

(R1) *The RKHS $\mathcal{H}$ is rich enough that $\inf_{f \in \mathcal{H}} \|f - f^*\|_{L^2} = 0$;*

(R2) *The cross-validation procedure (16) is asymptotically oracle-efficient in the sense that $\mathrm{CV}(\hat{\boldsymbol{\lambda}}) - \inf_{\boldsymbol{\lambda}} \mathrm{CV}(\boldsymbol{\lambda}) = o_p(1)$;*

(R3) *The penalty C_j is continuous in $\|\cdot\|_{L^2}$ and the parameter space for μ_j includes zero;*

the cross-validated multiplier satisfies $\hat{\mu}_j \xrightarrow{p} 0$ as $n \to \infty$.

Proof. The proof is given in Appendix A. □

Remark 23 (Practical implication)*.* Proposition 22 provides a theoretical guarantee that misspecified theories "turn themselves off" through cross-validation. In finite samples, a moderately misspecified theory may still receive positive $\hat{\mu}_j$ if it captures useful structure despite not being exactly correct. The estimator thus exploits partial truths—a theory need not be exactly right to be useful for prediction.

Remark 24 (Misspecification and variable coverage)*.* Consider a concrete case: the investment CAPM (Restriction 14) predicts expected returns as a function of investment-to-capital ratio and profitability, but says nothing about consumption growth. Does this mean the penalty implicitly imposes that consumption has zero effect? No. Each C_j constrains f along the specific dimensions of $\mathbf{X}$ that enter the model's predictions, but leaves f's dependence on other variables entirely unrestricted. The investment penalty pushes $f(\mathbf{X}_s)$ toward the production model's prediction $g_j(I/K, \mathrm{ROE})$ at each observation s, but f remains free to use consumption growth, sentiment, or any other variable in $\mathbf{X}$ to satisfy that target. Silence about a variable is not the same as imposing it has zero effect. The

combined penalty $\sum_j \mu_j C_j(f)$ covers many dimensions of the characteristic space through different restrictions, and the $\mu_j \to 0$ mechanism (Proposition 22) handles restrictions that are quantitatively wrong, not merely incomplete.

Remark 25 (Comparison to Bayesian model averaging). The misspecification robustness of theory-informed KRR is analogous to Bayesian model averaging (BMA) in spirit: both frameworks downweight models that fit poorly out of sample. However, the mechanism differs. BMA assigns posterior probabilities to discrete model specifications, whereas theory-informed KRR treats each model's restrictions as a continuous penalty with cross-validated strength. The latter avoids the combinatorial explosion of the BMA model space and allows partial incorporation of each theory's restrictions.

Remark 26 (Structural penalties and the saturation barrier). Standard KRR is minimax-optimal when the target function f^* satisfies a source condition $f^* \in [\mathcal{H}]^s$ with $0 < s \leq 1$, but saturates for smoother targets: when $s > 1$, the convergence rate cannot improve beyond $n^{-2/(2+\beta)}$ regardless of how λ_{stat} is tuned, where β governs the eigenvalue decay of the kernel integral operator (Li and Zhang, 2023; Zhang et al., 2025). This saturation effect is intrinsic to all spectral regularization methods, which lack the feature-learning capability of adaptive estimators (Barzilai et al., 2024). The structural penalties $\sum_j \mu_j C_j(f)$ can partially mitigate saturation by reducing the effective function class from the full RKHS $\mathcal{H}$ to a smaller set consistent with economic theory. When the penalties correctly narrow the search space, the effective source condition improves and the saturation boundary shifts upward. In the geometrical framework of Zhang et al. (2024a), this corresponds to projecting the estimator onto a lower-dimensional subspace of $\mathcal{H}$ compatible with the active structural restrictions, reducing the effective estimation dimension and improving the bias-variance trade-off.

2.9 Inference on Model Value

Beyond prediction, one may wish to test whether a particular economic model contributes statistically significant information beyond agnostic regularization. This amounts to testing

$$H_0 : \mu_j^* = 0 \qquad \text{vs.} \qquad H_1 : \mu_j^* > 0, \tag{17}$$

where μ_j^* is the population-optimal multiplier for model j. Rejection of H_0 implies that model j provides marginal predictive value beyond what the kernel and other models already capture.

2.9.1 Connection to Uniform Inference for KRR

Recent work by Singh and Vijaykumar (2023) establishes uniform confidence bands for kernel ridge regression estimators of conditional expectations. Their framework provides pointwise and uniform inference for $f^*(\mathbf{x})$ across $\mathbf{x} \in \mathcal{X}$ using a bias-corrected estimator and variance estimation based on the kernel's spectral properties.

We outline how this can be extended to test (17).

Proposition 27 (Testing the marginal value of economic models)**.** *Suppose the conditions of Singh and Vijaykumar (2023) hold for the RKHS $\mathcal{H}$, augmented with the following:*

(T1) *The mapping $\mu_j \mapsto \hat{f}(\cdot\,;\mu_j)$ is differentiable in L^2, with derivative $\partial_{\mu_j}\hat{f}$ satisfying $\|\partial_{\mu_j}\hat{f}\|_{L^2} = O(1)$;*

(T2) *The estimation error satisfies $\|\hat{f}(\cdot\,;\hat{\boldsymbol{\lambda}}) - f^*\|_\infty = O_p(r_n)$ for a rate $r_n \to 0$.*

Define the test statistic

$$T_j \;=\; n \cdot \big[\mathrm{CV}(\hat{\boldsymbol{\lambda}}_{-j}) - \mathrm{CV}(\hat{\boldsymbol{\lambda}})\big], \tag{18}$$

where $\hat{\boldsymbol{\lambda}}_{-j}$ denotes the cross-validated hyperparameters with μ_j constrained to zero. Under $H_0\colon \mu_j^* = 0$, $T_j = o_p(1)$. Under $H_1\colon \mu_j^* > 0$, $T_j \to \infty$ in probability.

Proof. The proof is given in Appendix A. □

Remark 28 (Permutation inference). A full distributional theory for T_j under H_0 (critical values, size control, power analysis) requires characterizing the joint distribution of the cross-validated hyperparameters, which we defer to future work. In practice, we implement permutation-based inference: randomly reassign the structural penalty C_j to pseudo-models (scrambling the economic content while preserving the penalty's functional form) and compare T_j to the permutation distribution. This yields valid p-values under exchangeability of the pseudo-model assignments.

Remark 29 (Asymptotic foundations for μ_j inference). The recent results of Fischer and Steinwart (2024) provide a rigorous asymptotic theory for linear functionals of KRR estimators. For any functional $\ell(f) = \langle f, g\rangle_{\mathcal{H}}$ with $g \in \mathcal{H}$, they establish a central limit theorem: the standardized functional converges to $\mathcal{N}(0,1)$ under appropriate regularization. Point evaluations achieve variance $O(n^{-1}\lambda^{-d/m})$ and bias $O(\lambda^{1-d/(2m)})$, where d is the input dimension and m the kernel smoothness order. The optimal $\lambda \sim n^{-1}$ yields a parametric rate for sufficiently smooth functionals. Since the gradient of C_j evaluated at $\hat{f}$ is a linear functional of $\hat{f}$, these results support a score-type test for $H_0 : \mu_j = 0$: evaluate the derivative $\partial C_j(f)/\partial f$ at the unconstrained KRR solution $\hat{f}^{\mathrm{KRR}}$, and reject when this gradient is large relative to its asymptotic standard error. This provides an alternative to the permutation-based approach of Remark 28, grounded in distributional theory rather than resampling. For uniform inference across evaluation points, Singh (2024) constructs valid confidence bands using a Gaussian multiplier bootstrap calibrated to the RKHS norm.

Remark 30 (Interpretation). The test (17) asks a fundamentally different question from testing whether a model's pricing errors are zero. A model may have large pricing errors ($C_j(f^*) > 0$) yet still receive $\mu_j^* > 0$ if its restrictions are correlated with the true f^* in a way that improves finite-sample estimation. Conversely, a correctly specified model ($C_j(f^*) = 0$) may have $\mu_j^* = 0$ if the RKHS and data are rich enough that the model's restrictions are redundant. The test thus measures the *marginal predictive value* of each theory in the

presence of all others.

Remark 31 (Relation to model confidence sets). The pairwise comparison underlying (18) connects to the model confidence set (MCS) framework of Hansen, Lunde, and Nason (2011). One could construct an MCS of theories whose marginal contributions are statistically indistinguishable: $\widehat{\mathrm{MCS}}_\alpha = \{j : T_j \leq c_\alpha\}$, where c_α is calibrated by the permutation distribution. This provides a principled way to report which theories are "useful" for asset-pricing prediction at a given confidence level.

3 Structural Restrictions Catalog

Each restriction in our framework originates from a specific economic model and imposes a distinct functional constraint on the mapping from observable characteristics to expected returns. A consumption-based Euler equation with power utility implies a different penalty $C_j(f)$ than one with Epstein–Zin preferences, even though both restrict the same SDF moment condition, because the functional form of the stochastic discount factor $M_{j,t+1}$ differs. Similarly, a production-based investment CAPM penalty differs from a q-theory penalty not merely in its parameter values but in which investment variables enter the mapping g_j and how they interact. The penalties are not interchangeable: each encodes a qualitatively different economic hypothesis about why expected returns vary across assets and over time.

We organize the fifty-six restrictions into eight families, grouped by economic mechanism: consumption-based Euler equations, production-based and investment restrictions, intermediary and institutional restrictions, information and learning restrictions, demand-system and market microstructure restrictions, variance and volatility restrictions, behavioral and sentiment restrictions, and macro-finance and term structure restrictions. For each restriction, we specify three elements: the source model (the economic theory that implies it), the mathematical form of the restriction (the SDF, first-order condition, or equilibrium condition), and the penalty functional $C_j(f)$ through which it enters the Theory-KRR objective (2)

from Section 2. Because every penalty $C_j(f)$ depends on f only through its evaluations at observed data points $\{f(\mathbf{X}_s)\}_{s=1}^n$, the representer theorem (Proposition 11) applies to the full collection, and the estimator remains computationally tractable regardless of how many restrictions are active.

3.1 Consumption-Based Euler Equations

Consumption-based models derive asset-pricing restrictions from the representative agent's intertemporal optimality condition. The Euler equation $\mathbb{E}[M_{t+1}R_{i,t+1}^{\text{gross}} \mid \mathcal{F}_t] = 1$ must hold for every asset i and date t, where the SDF M_{t+1} is determined by the agent's preferences and the consumption process. Different preference specifications—power utility, recursive utility, habit formation, long-run risk—yield different SDFs and hence different penalties. All thirteen restrictions in this family are Type A (Euler equation) penalties as defined in Section 2.3.1:

$$C_j^{\text{Euler}}(f) \;=\; \sum_{t=1}^{T}\left(\frac{1}{N}\sum_{i=1}^{N} M_j\big(\hat{\theta}_j, \mathbf{X}_t, \mathbf{X}_{t+1}\big)\cdot\big(f(\mathbf{X}_{i,t}) + R_t^f\big) - 1\right)^2, \tag{19}$$

where $\hat{\theta}_j$ denotes pre-estimated preference parameters obtained by GMM. What distinguishes the thirteen restrictions is solely the form of M_j.

Table 3 lists the thirteen consumption-based restrictions. The thirteen consumption-based restrictions span several distinct economic channels. Restrictions 1–5 vary preferences while maintaining i.i.d. or simple Markov consumption dynamics; what differs is how marginal utility depends on consumption growth and, in the habit models, on the history of consumption. Power utility (Restriction 1) serves as the baseline, and each subsequent preference specification adds a mechanism—intertemporal substitution separated from risk aversion (Restriction 2), slow-moving external habit (Restriction 3), or direct dependence on lagged consumption (Restrictions 4–5). These five SDFs share the same data requirements (aggregate consumption growth) but impose different functional relationships between M_{t+1}

and Δc_{t+1}.

Restrictions 6–9 modify the consumption *process* rather than preferences alone. Long-run risk (Restriction 6) introduces a persistent expected-growth component x_t and stochastic volatility σ_t, both of which enter the SDF through Epstein–Zin recursion; the SDF depends on the news to x_t and σ_t^2, not merely on realized consumption growth. Adding jumps (Restriction 7) generates occasional large movements in volatility that are absent under Gaussian dynamics. Rare disaster models (Restrictions 8–9) replace continuous volatility variation with low-probability, high-impact consumption contractions. The economic content differs sharply: long-run risk explains the equity premium through intertemporal hedging demand, while disaster models explain it through the pricing of tail events.

Restrictions 10–13 extend the framework along further dimensions. Precautionary savings (Restriction 10) introduces higher-order risk terms via the prudence coefficient, generating SDF dependence on conditional consumption variance. Durable consumption (Restriction 11) adds a second consumption good with its own intratemporal elasticity. Heterogeneous agents (Restriction 12) replaces the representative consumer with a continuum of agents whose idiosyncratic consumption risk enters the aggregate SDF through the cross-sectional variance of consumption growth. The nonparametric Euler restriction (Restriction 13) abandons parametric preferences entirely, requiring only that some monotone decreasing function of consumption growth prices assets correctly—it serves as a benchmark that detects whether *any* consumption-based SDF is empirically useful, without committing to a specific functional form.

3.2 Production-Based and Investment Restrictions

Production-based asset-pricing models derive expected returns from firms' intertemporal investment decisions. The key insight is that firms equate the marginal cost of investment today to the expected discounted marginal benefit, and this first-order condition links expected stock returns to observable investment and profitability variables. All ten restrictions

in this family are Type B (production FOC) penalties as defined in Section 2.3.2:

$$C_j^{\text{prod}}(f) \;=\; \frac{1}{n}\sum_{s=1}^{n}\big(f(\mathbf{X}_s) - g_j(\mathbf{Z}_s^{(j)})\big)^2, \tag{20}$$

where g_j maps firm-level investment characteristics $\mathbf{Z}_{i,t}^{(j)}$ to expected returns. The functional form of g_j and the variables that enter $\mathbf{Z}_{i,t}^{(j)}$ vary across restrictions.

Table 4 lists the ten production-based restrictions. The production-based restrictions divide into three groups. Restrictions 14–18 concern the technology of physical capital adjustment. The investment CAPM (Restriction 14) and q-theory (Restriction 15) provide the broadest frameworks: expected returns are determined by firms' optimal investment behavior, with investment-to-capital ratio and profitability as the key state variables. Restrictions 16–18 refine the adjustment cost specification. Convex adjustment costs (Restriction 16) yield a smooth, monotone relationship between I/K and expected returns. Nonconvex adjustment costs (Restriction 17) and irreversibility (Restriction 18) break this smoothness: large adjustments incur fixed costs, and disinvestment is more expensive than investment. These restrictions generate different functional forms for g_j—Restriction 16 predicts a linear or quadratic relationship in I/K, while Restrictions 17–18 predict threshold effects and asymmetries.

Restrictions 19–21 address financial frictions. These models predict that expected returns reflect not only the marginal product of capital but also the shadow cost of financing constraints. The three restrictions differ in which friction binds: collateral constraints (Restriction 19) link the premium to the ratio of capital to debt, equity issuance costs (Restriction 20) link it to leverage and internal cash flow, and costly external finance (Restriction 21) links it to the dividend payout and leverage. All three predict that financially constrained firms earn higher expected returns, but the mapping from observables to the constraint's shadow value differs across models.

Restrictions 22–23 expand the set of productive inputs beyond physical capital. Labor

adjustment costs (Restriction 22) introduce the hiring rate as an additional state variable in the investment FOC, generating a two-dimensional mapping $(I/K, \mathrm{HN}) \mapsto \mathbb{E}[R]$. Intangible capital (Restriction 23) adds R&D spending and selling, general, and administrative expenses as measures of intangible investment, recognizing that knowledge capital and organizational capital are productive assets with their own adjustment costs.

3.3 Intermediary and Institutional Restrictions

Intermediary asset-pricing models posit that marginal investors are not households but financial intermediaries—broker-dealers, banks, hedge funds—whose pricing kernels reflect institutional constraints on leverage, capital, and funding. When intermediaries are the marginal pricers, expected returns depend on intermediary-specific state variables: capital ratios, leverage, funding spreads, and balance-sheet capacity. A linear factor model with factor $F_{j,t}$ implies the SDF $M_{j,t+1} = a_j + b_j F_{j,t+1}$, which is a special case of a linear SDF. All eight restrictions in this family are therefore Type A (Euler equation) penalties as defined in Section 2.3.1:

$$C_j^{\text{Euler}}(f) \;=\; \sum_{t=1}^{T} \left(\frac{1}{N} \sum_{i=1}^{N} \big(\hat{a}_j + \hat{b}_j\, F_{j,t+1}\big) \cdot \big(f(\mathbf{X}_{i,t}) + R_t^f\big) - 1 \right)^2, \tag{21}$$

where $F_{j,t}$ is the intermediary factor and $\hat{a}_j$, $\hat{b}_j$ are pre-estimated SDF parameters obtained by GMM. This formulation requires only the factor time series—no pre-estimated betas are needed.

Table 5 lists the eight intermediary restrictions. The intermediary restrictions separate into models of intermediary health (Restrictions 24–25, 27) and models of market frictions that constrain intermediation (Restrictions 26, 28–31). The intermediary capital ratio (Restriction 24) and leverage (Restriction 25) factors capture the state of broker-dealer balance sheets: when capital is depleted or leverage is high, the marginal value of wealth to the intermediary rises, increasing risk prices. The dealer balance sheet factor (Restriction 27)

constructs an equity capital ratio specifically for primary dealers, providing a more direct measure of the capacity constraint. These three restrictions use different empirical proxies for the same conceptual channel—intermediary distress—and their relative μ_j values reveal which proxy captures the pricing-relevant variation.

Restrictions 26 and 28–31 focus on frictions in the trading environment. Funding liquidity (Restriction 26) links expected returns to the cost and availability of short-term funding: when funding dries up, margins increase and leveraged positions are unwound, creating fire-sale dynamics. Limits to arbitrage (Restriction 28) and slow-moving capital (Restriction 29) introduce a role for mispricing persistence—arbitrageurs face constraints (capital, risk limits, organizational frictions) that prevent immediate price correction. The margin CAPM (Restriction 30) derives a two-beta model in which both market exposure and margin exposure are priced, while the leverage-constraint model (Restriction 31) predicts a compressed security market line because leverage-constrained investors overweight high-beta assets. The key distinction: Restrictions 24–27 model *who* the marginal pricer is, while Restrictions 28–31 model *what constrains* the marginal pricer.

3.4 Information and Learning Restrictions

Information and learning models generate expected-return variation from agents' incomplete knowledge about economic fundamentals. Unlike consumption-based models, where the SDF is pinned down by preferences and the consumption process, these models derive pricing implications from the structure of beliefs—how agents learn, what they are uncertain about, and how they process information. These restrictions generate Type A penalties: Restrictions 32–35 specify a complete SDF from the learning model, while Restrictions 36–37 imply a linear SDF driven by a belief-related state variable (disagreement or information flow), which reduces to the Euler equation form $M_{j,t+1} = a_j + b_j F_{j,t+1}$.

Table 6 lists the six information restrictions. The six information restrictions differ in *what* agents are uncertain about and *how* they process that uncertainty. Bayesian learning

(Restriction 32) generates time-varying risk premia because the posterior variance of growth declines as data accumulate, changing the conditional distribution of the SDF. Regime learning (Restriction 33) produces a distinct mechanism: agents infer a latent state (expansion vs. recession), and uncertainty about the current regime amplifies return volatility and risk premia—the SDF depends on the belief π_t rather than on a parameter estimate. Parameter uncertainty (Restriction 34) prices the risk that the true data-generating process is unknown, generating a premium for model risk that shrinks only slowly with sample size.

Ambiguity aversion (Restriction 35) departs from Bayesian updating altogether: the agent entertains a set of plausible models $\mathcal{P}$ and evaluates each asset under the worst-case model. The SDF is the likelihood ratio between the worst-case and reference models, generating pricing implications distinct from those of any single Bayesian model. Disagreement (Restriction 36) and rational inattention (Restriction 37) shift focus from a single agent's beliefs to the cross-section of beliefs. Disagreement models price the dispersion of beliefs as a risk factor (optimists hold long positions, pessimists short, and their trading generates a risk premium). Rational inattention models generate endogenous information asymmetry: agents with limited processing capacity allocate attention optimally, and the resulting heterogeneity in information sets produces cross-sectional variation in expected returns.

3.5 Demand-System and Market Microstructure Restrictions

Demand-system models derive asset prices from the aggregation of heterogeneous investors' portfolio demands. The equilibrium restriction is market clearing: aggregate demand for each asset must equal its supply. This yields Type D (demand-system equilibrium) penalties as defined in Section 2.3.3:

$$C_j^{\text{demand}}(f) \;=\; \sum_{t=1}^{T} \left\| \sum_i w_i^{(j)}\, D_i^{(j)}\big(\hat{\mathbf{f}}_t\big) - \mathbf{S}_t \right\|^2 , \tag{22}$$

where $\hat{\mathbf{f}}_t = (f(\mathbf{X}_{1,t}), \ldots, f(\mathbf{X}_{N,t}))^\top$ and $\mathbf{S}_t$ denotes market-capitalization shares. Restrictions 41–43 involve microstructure frictions and take modified forms as described below.

Table 7 lists the six demand-system and microstructure restrictions. The demand-system restrictions (38–40) share the market-clearing equilibrium condition but differ in the structure of investors' demand functions. Restriction 38 models demand as a function of observable characteristics and a latent demand shock, estimating the demand system directly from portfolio holdings data. Restriction 39 derives demand from investors' optimality conditions with characteristic-based demand elasticities, providing a micro-founded demand function. Restriction 40 focuses on the *aggregate* demand elasticity: when total market demand is inelastic, flow-driven trading has outsized price impact, and expected returns reflect the accommodation of demand shocks rather than fundamental risk.

Restrictions 41–43 address frictions that prevent instantaneous and costless trading. Search frictions (Restriction 41) apply primarily to over-the-counter markets, where finding a counterparty takes time; the expected-return penalty includes the search intensity and the buyer's bargaining power. Fire sale externalities (Restriction 42) generate a Type A Euler penalty with $M_{42,t+1} = \hat{a}_{42} + \hat{b}_{42}\,\mathrm{FSP}_{t+1}$: when multiple leveraged institutions simultaneously liquidate positions, the fire-sale pressure factor enters the SDF. Short-sale constraints (Restriction 43) restrict the adjustment mechanism on the negative side: when pessimists cannot short, asset prices reflect only optimists' valuations, and the degree of overpricing depends on the divergence of opinions and the binding intensity of the constraint.

3.6 Variance and Volatility Restrictions

Variance and volatility models link expected returns to measures of conditional variance and the compensation investors demand for bearing variance risk. These two restrictions use publicly available data—the CBOE Volatility Index (VIX) as a measure of implied variance and daily CRSP returns to compute realized variance—rather than individual equity option data, making them applicable to the broad cross-section over the post-1990 period. Both

restrictions generate Type B penalties that map volatility-related state variables to expected returns.

Table 8 lists the two variance and volatility restrictions. The two variance and volatility restrictions capture complementary aspects of volatility pricing. The variance risk premium (Restriction 44) uses the spread between implied variance (from VIX) and expected realized variance (computed from daily CRSP returns within each month) as a predictor of aggregate returns; the penalty penalizes discrepancies between $f(\mathbf{X}_t)$ and the VRP-implied expected return. Bollerslev et al. (2009) show that VRP is a strong predictor of market returns at horizons from one to six months, and the restriction encodes this predictive relationship. Stochastic volatility (Restriction 45) uses the conditional variance V_t from a Heston-type model, where VIX serves as a proxy for the latent conditional variance and the variance process carries its own risk premium distinct from the equity premium. Together, these restrictions ensure that the estimator respects the well-documented relationship between variance dynamics and expected returns without requiring granular option-level data.

3.7 Behavioral and Sentiment Restrictions

Behavioral models derive expected-return patterns from departures from rational expectations or expected utility. These restrictions do not derive an SDF from a representative agent's Euler equation; instead, they predict cross-sectional or time-series return patterns as functions of behavioral variables—past returns, capital gains, sentiment indices. Most generate Type B penalties, where g_j maps behavioral state variables to expected returns.

Table 9 lists the six behavioral restrictions. The behavioral restrictions exploit distinct psychological mechanisms. Prospect theory (Restriction 46) modifies the SDF with a kink at the reference point: losses are weighted more heavily than gains by a factor $\lambda > 1$, generating a first-order risk premium for assets whose returns covary with movements around the reference point. This is the only behavioral restriction that directly specifies an SDF, placing it closest to the consumption-based framework.

Restrictions 47–49 generate cross-sectional return predictability from belief formation biases. Overconfidence (Restriction 47) produces momentum at short horizons (agents overreact to private signals, pushing prices beyond fundamentals) and reversal at long horizons (prices revert as public information arrives). The disposition effect (Restriction 48) predicts return continuation among stocks with large unrealized gains and return reversal among stocks with large unrealized losses, because investors' reluctance to realize losses impedes price adjustment. Extrapolative expectations (Restriction 49) model agents who weight recent returns too heavily when forming expectations, generating a specific functional form for the relationship between past returns and expected future returns. These three restrictions share momentum-related variables as inputs but predict different functional relationships between past returns and expected future returns.

Investor sentiment (Restriction 50) operates at the aggregate level: the Baker–Wurgler sentiment index predicts that, conditional on high sentiment, speculative stocks (small, unprofitable, high-volatility) earn low subsequent returns, while safe stocks are largely unaffected. This generates a Type A Euler equation penalty where $M_{50,t+1} = \hat{a}_{50} + \hat{b}_{50}\,\mathrm{Sent}_{t+1}$ is a linear SDF in the sentiment factor. Salience theory (Restriction 51) predicts mispricing at the individual asset level: stocks with salient (extreme) upside payoffs are overpriced because agents overweight the probability of salient outcomes, generating a negative relationship between salience and expected returns.

3.8 Macro-Finance and Term Structure Restrictions

Macro-finance models link expected returns on equities and bonds to macroeconomic state variables: the yield curve, inflation expectations, monetary policy, and consumption-wealth ratios. These restrictions enter as Type A or Type B penalties depending on whether the model implies a linear SDF driven by a macro factor (Type A) or a direct functional mapping from macro variables to expected returns (Type B).

Table 10 lists the five macro-finance restrictions. The macro-finance restrictions link

equity expected returns to macroeconomic and fixed-income state variables through five distinct channels. The affine term structure restriction (Restriction 52) imposes no-arbitrage across maturities, using the yield curve state vector $\mathbf{y}_t$ (level, slope, curvature, and potentially macro factors) to constrain the time-varying equity risk premium. This restriction is structurally different from the others because it derives from the bond market and imposes cross-market no-arbitrage.

Restrictions 53–54 transmit monetary and inflation shocks to equity prices. The inflation risk premium (Restriction 53) uses the spread between nominal and real yields to extract the compensation investors demand for bearing inflation uncertainty; this premium predicts equity returns because inflation uncertainty covaries with real economic risk. The monetary policy restriction (Restriction 54) identifies the surprise component of federal funds rate decisions (using fed funds futures) and maps it to equity returns, exploiting the direct channel from policy rates to discount rates.

Restrictions 55 and 56 use slow-moving macroeconomic aggregates. The consumption-wealth ratio *cay* (Restriction 55) exploits cointegration among consumption, asset wealth, and labor income: when consumption is high relative to wealth, expected returns are high because the ratio mean-reverts through subsequent returns. The excess bond premium (Restriction 56) isolates the component of credit spreads that reflects risk appetite and financial intermediary health, net of default risk; it predicts equity returns because deteriorating credit conditions signal tightening financial constraints economy-wide.

3.9 Summary and Discussion

The catalog comprises fifty-six restrictions organized into eight families. Each restriction imposes a distinct functional form on the mapping from observable characteristics $\mathbf{X}_{i,t}$ to expected returns $f(\mathbf{X}_{i,t})$, and each translates into a specific penalty functional $C_j(f)$ within the Theory-KRR objective (2). The penalties are heterogeneous in structure: Type A penalties enforce Euler equation moment conditions and require specifying the SDF—either from

a structural model or from a linear factor representation $M_{t+1} = a + bF_{t+1}$ (Restrictions 1–13, 24–37, 42, 46, 50, 53, 56); Type B penalties equate expected returns to functions of investment, behavioral, or macro variables through first-order conditions (Restrictions 14–23, 41, 43, 44–45, 47–49, 51–52, 54–55); and Type D penalties enforce market-clearing equilibrium conditions from demand-system models (Restrictions 38–40). Despite this heterogeneity, the framework in Section 2 accommodates all fifty-six penalties within a single objective: each $C_j(f)$ depends on f only through its evaluations at observed data points, the representer theorem applies, and cross-validation selects the vector $\boldsymbol{\mu}$ that balances predictive fit against theoretical guidance.

A property of the catalog that makes the μ_j ranking meaningful is that many restrictions share observable input variables but impose different functional relationships. Consumption growth enters Restrictions 1–13 through thirteen different SDFs. Investment-to-capital ratio enters Restrictions 14–18 through five different adjustment cost specifications. Past returns enter Restrictions 47–49 through three different behavioral mechanisms. When two restrictions share inputs but one receives a higher μ_j in cross-validation, the data have identified which functional relationship between those inputs and expected returns is more empirically useful. This is the sense in which the cross-validated multipliers $\boldsymbol{\mu}$ produce a ranking of economic theories: they rank not just which state variables matter, but which *economic mechanisms linking state variables to expected returns* are supported by the data.

4 Estimation and Implementation

This section addresses the practical aspects of taking Theory-KRR from the mathematical framework of Sections 2–3 to an implementable estimator. We describe the kernel choice and bandwidth calibration (Section 4.1), the pre-estimation of structural parameters that enter the penalty functionals (Section 4.2), the cross-validation strategy for selecting the high-dimensional vector of penalty multipliers (Section 4.3), computational details (Section 4.4),

and the rolling-window procedure for out-of-sample evaluation (Section 4.5).

4.1 Kernel Choice and Bandwidth Selection

The kernel $k : \mathbb{R}^p \times \mathbb{R}^p \to \mathbb{R}$ determines the RKHS $\mathcal{H}$ in which the estimator searches for the conditional expectation function f^*. Our primary specification is the Gaussian radial basis function (RBF) kernel:

$$k(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\sigma^2}\right), \tag{23}$$

where $\sigma > 0$ is the bandwidth parameter and $\|\cdot\|$ denotes the Euclidean norm in $\mathbb{R}^p$. The Gaussian kernel is universal: the associated RKHS is dense in the space of continuous functions on any compact subset of $\mathbb{R}^p$, so the approximation error $\inf_{f \in \mathcal{H}} \|f - f^*\|$ can be made arbitrarily small. This universality is essential for our application, where the true conditional expectation function may exhibit the nonlinearities, interactions, and threshold effects documented in the cross-section of returns.

We set the bandwidth σ using the median heuristic:

$$\sigma = \text{median}\{\|\mathbf{X}_{i,t} - \mathbf{X}_{j,s}\| : (i,t) \neq (j,s)\}. \tag{24}$$

The median heuristic places the kernel bandwidth at a scale where roughly half of the pairwise distances produce kernel evaluations above $e^{-1/2} \approx 0.61$ and half below. This data-driven rule avoids both extremes: a bandwidth that is too small relative to the data's spread produces a nearly diagonal kernel matrix (overfitting), while a bandwidth that is too large produces a nearly constant kernel matrix (underfitting, approaching the global mean). The median heuristic has strong theoretical support in the kernel methods literature and performs well in high-dimensional settings where manual tuning is infeasible.

Two limiting cases clarify the role of the kernel. A linear kernel $k(\mathbf{x}, \mathbf{x}') = \mathbf{x}^\top \mathbf{x}'$ restricts $\mathcal{H}$ to linear functions $f(\mathbf{x}) = \boldsymbol{\beta}^\top \mathbf{x}$, and Theory-KRR reduces to a penalized linear regression with structural penalties on coefficients. This nesting (Proposition 19(iii)) allows us to

isolate the contribution of nonlinearity from the contribution of theory-informed penalties. At the other extreme, the Gaussian kernel with bandwidth $\sigma \to 0$ produces a kernel matrix approaching the identity, so $\hat{f}$ interpolates the data subject only to the structural penalties. The joint configuration of σ, λ_{stat}, and $\boldsymbol{\mu}$ determines the estimator's regularization, and cross-validation selects the balance that minimizes OOS prediction error.

4.2 Estimation of Structural Parameters

Each structural penalty $C_j(f)$ depends on model-specific parameters θ_j that characterize the underlying economic model. For consumption-based models, θ_j includes the coefficient of relative risk aversion γ, the elasticity of intertemporal substitution ψ, the subjective discount factor δ, and the habit persistence parameter b. For production-based models, θ_j governs the curvature of the adjustment cost function. For intermediary models, θ_j parameterizes the SDF loadings on the intermediary factor. We consider three approaches to estimating θ_j, ordered from most to least principled.

Approach 1: Joint estimation (preferred). The most principled approach estimates f and $\{\theta_j\}_{j=1}^{J}$ simultaneously:

$$(\hat{f}, \hat{\theta}_1, \ldots, \hat{\theta}_J) \;=\; \underset{f \in \mathcal{H},\, \theta_1, \ldots, \theta_J}{\arg\min} \; \frac{1}{n} \sum_s (R_s - f(\mathbf{X}_s))^2 + \lambda_{\text{stat}} \|f\|_{\mathcal{H}}^2 + \sum_j \mu_j \, C_j(f; \theta_j). \tag{25}$$

This optimization is convex in f (given θ) and generally nonconvex in θ, but alternating minimization—updating f given θ and then θ given f—converges reliably from the pre-estimation starting point. Joint estimation is the most principled approach because it allows the structural parameters and the nonparametric function to adapt to each other: a consumption-based SDF with risk aversion γ that is optimal in isolation may not be optimal when combined with intermediary and production penalties.

Approach 2: Pre-estimation (two-stage). For computational simplicity, one can pre-estimate $\hat{\theta}_j$ in a first stage that is separate from the KRR optimization. For each consumption-based SDF model j, we estimate

$$\hat{\theta}_j = \arg\min_{\theta_j} \left\| \frac{1}{T} \sum_{t=1}^{T} \mathbf{z}_t \left[M_j(\theta_j, \mathbf{X}_t, \mathbf{X}_{t+1}) \, R^{\text{gross}}_{m,t+1} - 1 \right] \right\|^2_{\hat{\mathbf{W}}^{-1}}, \tag{26}$$

where $\mathbf{z}_t$ is a vector of instruments (lagged consumption growth, the dividend-price ratio, the term spread) and $\hat{\mathbf{W}}$ is an HAC-consistent estimate of the instrument covariance matrix. For production-based models, we estimate adjustment cost parameters from firm-level investment regressions. For intermediary models, the SDF loadings $(\hat{a}_j, \hat{b}_j)$ are estimated by GMM on the Euler equation $\mathbb{E}[M_j R^{\text{gross}}] = 1$ using test assets. The pre-estimated $\hat{\theta}_j$ is then plugged into $C_j(f)$ and treated as fixed during the second-stage KRR optimization. This approach ignores the interaction between f and θ: the structural parameters estimated from aggregate moments may not be optimal for cross-sectional return prediction.

Approach 3: Calibration. For parameters with well-established values, we adopt published calibrations from the original papers. We set $\delta = 0.99$ (monthly) in all consumption-based models and use $b = 0.87$ from Campbell and Cochrane (1999) for the external habit model. This is the simplest approach: $C_j(f)$ is fully determined by the data and a known parameter vector, with no estimation of θ_j. The risk is that published parameter values, calibrated to match aggregate moments in a specific sample, may not be appropriate for cross-sectional return prediction.

Our baseline uses joint estimation (Approach 1), initialized from the pre-estimation values (Approach 2). Robustness analysis compares all three approaches. For the sensitivity analysis, we report results over a grid of values for the most influential parameters:

$$\gamma \in \{2, 5, 10, 20\}, \qquad \delta \in \{0.95, 0.97, 0.99\}. \tag{27}$$

This grid spans the range of calibrations used in the asset-pricing literature, from low-risk-aversion economies ($\gamma = 2$) to the high-risk-aversion calibrations required by rare disaster models ($\gamma = 20$).

4.3 Cross-Validation Strategy

The full hyperparameter vector of Theory-KRR is $(\lambda_{\text{stat}}, \mu_1, \ldots, \mu_J)$, where $J = 56$ in our baseline specification. A 57-dimensional grid search is computationally infeasible. We reduce dimensionality by exploiting the economic structure of the restrictions and adopt the three-way temporal split of Gu et al. (2020) (Section 1.1) for hyperparameter selection.

We group the 56 restrictions into $G = 8$ families corresponding to the eight economic model classes cataloged in Section 3: consumption-based Euler equations, production-based restrictions, intermediary restrictions, information and learning restrictions, demand-system restrictions, variance and volatility restrictions, behavioral restrictions, and macro-finance restrictions. Within each family g, we impose a common multiplier:

$$\mu_j = \mu_{g(j)}^{\text{group}} \qquad \text{for all } j \text{ in family } g. \tag{28}$$

This reduces the search space from 61 dimensions to 9: one statistical penalty λ_{stat} and eight group-level multipliers $\mu_1^{\text{group}}, \ldots, \mu_8^{\text{group}}$. The grouping is motivated by the observation that restrictions within the same economic family share a common source of identification and are likely to have correlated empirical content. The grouping is not restrictive in the dimension that matters: cross-validation can set any group's multiplier to zero, shutting down that entire family of restrictions.

Following Gu et al. (2020), we adopt a rolling three-way sample split. For each test year τ from 1987 to 2023, we define:

1. **Training sample** (July 1963 through year $\tau - 13$): estimate $f(\cdot)$ and all structural parameters $\hat{\theta}_j$ for each candidate hyperparameter configuration.

2. **Validation sample** (year $\tau - 12$ through year $\tau - 1$, 12 years): select the hyperparameter vector $(\hat{\lambda}_{\text{stat}}, \hat{\boldsymbol{\mu}}^{\text{group}})$ that minimizes average squared prediction error:

$$(\hat{\lambda}_{\text{stat}}, \hat{\boldsymbol{\mu}}^{\text{group}}) = \underset{\lambda_{\text{stat}}, \boldsymbol{\mu}^{\text{group}}}{\arg\min} \frac{1}{n_{\text{val}}} \sum_{(i,t) \in \mathcal{S}_{\text{val}}} \left(R_{i,t+1} - \hat{f}_{\text{train}}(\mathbf{X}_{i,t}) \right)^2, \tag{29}$$

   where $\hat{f}_{\text{train}}$ is estimated using training data only.

3. **Testing sample** (year τ, 12 months): evaluate out-of-sample performance using the selected hyperparameters. This sample is never used for estimation or tuning.

For the first test year $\tau = 1987$, the training sample spans 1963–1974 (12 years) and the validation sample spans 1975–1986 (12 years). As τ advances, both the training and validation windows expand and slide forward: by 2023, the training sample spans 1963–2010 and the validation sample spans 2011–2022. The 12-year validation window follows Gu et al. (2020) and provides sufficient temporal variation for reliable hyperparameter selection. After selecting $(\hat{\lambda}_{\text{stat}}, \hat{\boldsymbol{\mu}}^{\text{group}})$ on the validation sample, we re-estimate f on the combined training and validation data (all months through year $\tau - 1$) and predict returns for year τ. This rolling protocol yields 37 annual out-of-sample predictions spanning 1987–2023, covering the 1987 crash, the dot-com bust, the global financial crisis, and the COVID-19 pandemic.

We search the 9-dimensional hyperparameter space using coordinate descent over the group multipliers, an approach inspired by the cyclic coordinate descent algorithm that underlies the Lasso and elastic net solvers of Friedman et al. (2010). The statistical penalty λ_{stat} is first selected by standard KRR cross-validation without structural penalties ($\mu_g = 0$ for all g). Holding λ_{stat} fixed, we then optimize the eight group multipliers by cycling through each group $g = 1, \ldots, G$ and performing a one-dimensional line search over a grid of candidate values:

$$\mu_g^{\text{group}} \in \{0\} \cup \{10^u : u \in \text{linspace}(-4, 1, 15)\}. \tag{30}$$

For each candidate value of μ_g^{group}, all other multipliers are held at their current values, and the validation-set MSE is evaluated. The value that yields the lowest MSE is retained, and the algorithm moves to the next group. One full pass through all $G = 8$ groups constitutes one cycle; we repeat for up to 3 cycles and stop early if no group multiplier changes in a complete cycle.

The total number of function evaluations is at most $3 \times 8 \times 16 = 384$, compared to k^9 for an exhaustive grid. Each evaluation requires solving one Nyström-approximated KRR system (an $m \times m$ Cholesky factorization with $m = 500$; see Appendix B.1), followed by a single Newton-like correction step that incorporates the structural penalty gradient. Because the 16 candidate values within each group are independent, they are evaluated in parallel across CPU cores, so each group's line search completes in the wall-clock time of a single evaluation. On our 40-core workstation, the full coordinate descent completes in under 5 seconds per rolling window.

This approach connects to the penalized GMM literature, where sequential moment selection achieves rates comparable to the oracle that knows which moments are valid (Caner, 2009; Cheng and Liao, 2015). Coordinate descent also has a natural economic interpretation: it asks, for each theory family in turn, "what is the optimal strength of this family's penalty, given the penalties already imposed by other families?" The sequential nature of the algorithm means that substitution effects between families—for instance, consumption-based and intermediary restrictions that capture similar risk exposures—are resolved automatically: the family that provides the largest marginal improvement is retained at a high multiplier, while the redundant family's multiplier is driven toward zero.

4.4 Computational Details

By Proposition 11, the estimator $\hat{f}(\cdot) = \sum_{s=1}^{n} \hat{\alpha}_s \, k(\mathbf{X}_s, \cdot)$ is parameterized by the coefficient vector $\hat{\boldsymbol{\alpha}} \in \mathbb{R}^n$. The optimization problem (8) is therefore finite-dimensional: we optimize over $\boldsymbol{\alpha}$ rather than over functions in $\mathcal{H}$.

When all active penalties are quadratic in the fitted values $\hat{\mathbf{f}} = \mathbf{K}\boldsymbol{\alpha}$, the objective is a convex quadratic in $\boldsymbol{\alpha}$, and the solution is given by the linear system (15). The dominant computational cost is the $O(n^3)$ matrix inversion (or, equivalently, the Cholesky decomposition of the $n \times n$ system matrix). We solve the system using Cholesky factorization, which is numerically stable and exploits the positive definiteness guaranteed by $\lambda_{\text{stat}} > 0$.

For problems involving Type A (Euler equation) penalties, the system matrix in (14) contains the term $\boldsymbol{\Omega}\,\mathbf{K}$, where $\boldsymbol{\Omega}$ includes the $\mathbf{A}_j^\top \mathbf{A}_j$ matrices. These are at most rank T (since $\mathbf{A}_j$ has T rows), and we exploit this low-rank structure by precomputing and caching the products $\mathbf{A}_j\,\mathbf{K}$ for each active Euler-equation restriction.

When structural penalties are active, we approximate the theory-informed solution by a single Newton-like correction from the standard KRR warm start. Specifically, let $\hat{\boldsymbol{\beta}}_{\text{KRR}}$ denote the Nyström KRR solution (all $\mu_j = 0$), and let $\mathbf{g} = \sum_j \mu_j\, \mathbf{Z}^\top \frac{\partial C_j}{\partial \hat{\mathbf{f}}}\big|_{\hat{\mathbf{f}}=\mathbf{Z}\hat{\boldsymbol{\beta}}_{\text{KRR}}}$ be the projected structural penalty gradient evaluated at the KRR fit. The theory-informed coefficient vector is

$$\hat{\boldsymbol{\beta}}_{\text{theory}} = \hat{\boldsymbol{\beta}}_{\text{KRR}} - \alpha\,(\mathbf{Z}^\top\mathbf{Z} + n\lambda_{\text{stat}}\,\mathbf{I}_m)^{-1}\mathbf{g}, \tag{31}$$

where the step size α is selected from a discrete set $\{0, 0.01, 0.05, 0.1, 0.5, 1, 2\}$ by minimizing the validation-set MSE. This one-step correction is computationally cheap—it requires one $m \times m$ Cholesky factorization plus one matrix-vector product per active restriction—and avoids the overhead of iterative L-BFGS optimization. Since the structural penalties are small relative to the data-fit term (by construction, as the μ_j are tuned to improve validation MSE rather than to enforce exact compliance), the single Newton step provides a good approximation to the fully converged solution.

The kernel matrix $\mathbf{K}$ is $n \times n$ and must be stored in memory. For our primary specification with $n \approx 4{,}000$ (cross-section of managed portfolios over 12 months), $\mathbf{K}$ occupies approximately 128 MB in double precision, which is manageable on standard hardware. For the full individual-stock panel ($n > 100{,}000$), storing $\mathbf{K}$ requires over 80 GB, which exceeds typical workstation memory.

Two standard approximations address this scaling challenge. The Nyström approximation selects a subset of $m \ll n$ landmark points and approximates $\mathbf{K} \approx \mathbf{K}_{nm}\mathbf{K}_{mm}^{-1}\mathbf{K}_{nm}^{\top}$, where $\mathbf{K}_{nm}$ is the $n \times m$ matrix of kernel evaluations between all points and the landmarks, and $\mathbf{K}_{mm}$ is the $m \times m$ kernel matrix among landmarks. This reduces memory to $O(nm)$ and solve time to $O(nm^2)$. Random Fourier features offer an alternative: they approximate the Gaussian kernel by an explicit feature map $\boldsymbol{\phi} : \mathbb{R}^p \to \mathbb{R}^D$ such that $k(\mathbf{x}, \mathbf{x}') \approx \boldsymbol{\phi}(\mathbf{x})^{\top}\boldsymbol{\phi}(\mathbf{x}')$, converting the kernel method into a linear regression in D-dimensional feature space. Both approximations introduce a controllable approximation error that decreases as m or D increases. We use $m = 500$ Nyström landmarks when the full kernel matrix is infeasible.

4.5 Rolling-Window Procedure

The rolling three-way split described in Section 4.3 defines the out-of-sample evaluation framework. At each annual rebalancing date (December of year $\tau - 1$), the full procedure is: (i) estimate $\hat{f}$ on the training sample using each candidate $(\lambda_{\text{stat}}, \boldsymbol{\mu}^{\text{group}})$, (ii) select the hyperparameter vector that minimizes validation-sample MSE, (iii) re-estimate $\hat{f}$ on the combined training and validation data (all months through December of year $\tau - 1$), and (iv) predict returns for the 12 months of year τ. Portfolio weights are set at the rebalancing date and held fixed for 12 months.

The expanding training window grows from 12 years (for $\tau = 1987$) to 48 years (for $\tau = 2023$), allowing the estimator to learn from the full history of cross-sectional return patterns. Annual rebalancing reduces computational cost (one full estimation per year rather than per month) and aligns with the holding-period assumptions of many characteristic-based strategies.

The hyperparameter vector $(\hat{\lambda}_{\text{stat}}, \hat{\boldsymbol{\mu}}^{\text{group}})$ is re-estimated at each rebalancing date, allowing the optimal balance between statistical and theory-informed regularization to vary over time. Tracking $\hat{\mu}_g^{\text{group}}$ across rebalancing dates reveals temporal variation in the value of each theory family—for instance, whether consumption-based restrictions become more

useful during recessions or whether intermediary restrictions gain importance after financial crises. We report the time series of estimated multipliers in Section 6.

All information used in estimation—returns, characteristics, macro state variables, and the structural parameters $\hat{\theta}_j$—is available as of the training window's end date. Characteristics are lagged by at least one month relative to the return being predicted. Macro variables use the vintage available at the prediction date, not subsequently revised values. The structural parameters $\hat{\theta}_j$ are pre-estimated using only data within the training window.

5 Data and Baseline Models

5.1 Data

Our primary dataset consists of monthly stock returns from the Center for Research in Security Prices (CRSP) monthly stock file. The sample includes ordinary common shares (share codes 10 and 11) listed on NYSE, AMEX, and NASDAQ (exchange codes 1, 2, and 3). Excess returns are computed by subtracting the one-month Treasury bill rate from the Fama-French data library. The effective sample period begins in July 1963—when both CRSP and Compustat coverage become reliable—and ends in December 2023, spanning 726 months and approximately 18,000 unique securities. This start date matches Gu et al. (2020) and the bulk of the cross-sectional return prediction literature.

We pair returns with 220 firm-level predictive characteristics constructed from CRSP returns, Compustat fundamentals, and the Open Source Asset Pricing (OSAP) dataset of Jensen et al. (2023). The OSAP dataset provides a standardized, publicly available panel of signed characteristics at the security-month level, constructed following the original papers' methodologies and validated against published results. Supplementary variables include CRSP-derived signals (price, size, short-term reversal) and Compustat-derived fundamentals (investment-to-capital, ROE, leverage, ROA, gross profitability, asset growth, R&D intensity) that enter the structural restriction penalties directly. All characteristics are signed

so that higher values predict higher expected returns. This characteristic set subsumes and extends the 94 characteristics used in Gu et al. (2020).

We merge the OSAP characteristics with CRSP returns using the permanent security identifier (permno). To ensure comparability across variables measured in different units, we cross-sectionally rank each characteristic to the unit interval $[0, 1]$ each month. Ranking is a monotone transformation that preserves the ordinal information in each characteristic while eliminating outliers and ensuring that all variables enter the kernel on a common scale.

The structural restrictions in Section 3 additionally require firm-level accounting data from Compustat. We obtain annual fundamentals (525,028 firm-years, 40,678 unique firms, 1950–2025) and quarterly fundamentals (1.87 million firm-quarters, 39,568 unique firms, 1961–2025) from WRDS. These are merged with CRSP via the CRSP-Compustat linking table using primary link types (LC and LU) with an effective link date filter. Key accounting variables include total assets, book equity, sales, income before extraordinary items, capital expenditures, property-plant-equipment, depreciation, long-term and short-term debt, cash holdings, R&D expenditure, and selling-general-administrative expenses. Our Compustat sample includes both active and inactive firms to avoid survivorship bias.

We augment firm-level characteristics with macroeconomic state variables that serve two roles: they enter the kernel as predictors of expected returns, and they supply inputs to the structural restriction penalties in Section 3. These variables are drawn from four sources.

From the Federal Reserve Economic Data (FRED), we obtain the 3-month Treasury bill rate (TB3MS), the 10-year Treasury yield (GS10), Moody's AAA and BAA corporate bond yields, the Consumer Price Index (CPIAUCSL), industrial production (INDPRO), and the unemployment rate (UNRATE). The term spread (GS10 minus TB3MS) and the default spread (BAA minus AAA) are computed as differences. From the National Income and Product Accounts via FRED, we obtain personal consumption expenditures on nondurable goods and services, the PCE price index, and population, which we combine to construct real per capita consumption growth—the central input to the thirteen consumption-based

Euler equation restrictions.

The intermediary capital ratio and intermediary capital risk factor of He et al. (2017) are obtained from the authors' website at quarterly frequency and interpolated to monthly. The consumption-wealth ratio *cay* of Lettau and Ludvigson (2001) is constructed following their Dynamic OLS methodology from FRED series on household net worth (TNWBSHNO), employee compensation (COE), and consumption, using 8 leads and lags of the differenced regressors. The Baker and Wurgler (2006) investor sentiment index (orthogonalized version) is obtained from the authors' data repository and covers 1965–2023. The CBOE Volatility Index (VIX) is obtained from FRED and is available from January 1990; we aggregate daily values to monthly averages. The excess bond premium of Gilchrist and Zakrajšek (2012) is proxied by the Moody's BAA spread over the 10-year Treasury (BAAFFM series from FRED). Additional macro predictors—the dividend-price ratio, earnings-price ratio, aggregate book-to-market, net equity expansion, Treasury bill rate, long-term yield, and stock variance—are obtained from the Welch and Goyal (2008) dataset, which covers 1871–2024 at monthly frequency.

Several structural restrictions require variables beyond the standard characteristic and macro sets. For the production-based restrictions (14–23), we construct leverage (total debt over total assets), R&D intensity (R&D expenditure over sales), SGA intensity (selling, general, and administrative expenses over total assets), and the collateral ratio (PP&E over total debt) from Compustat annual fundamentals. The hiring rate of Belo et al. (2014) is obtained from the OSAP characteristic panel. The dividend yield used in the costly external finance restriction (21) is the short-term dividend yield from the OSAP panel.

For intermediary restrictions, we supplement the HKM factor with broker-dealer leverage from the Federal Reserve's Financial Accounts (Flow of Funds series BOGZ1FL664090005Q and BOGZ1FL665080003Q), following Adrian et al. (2014). The TED spread (3-month LIBOR minus 3-month Treasury bill, FRED series TEDRATE) proxies funding liquidity in restriction 26; this series ends in January 2022 following the discontinuation of LIBOR.

The betting-against-beta (BAB) factor of Frazzini and Pedersen (2014) is obtained from AQR's data library and covers 1930–2025 at monthly frequency. Household equity net flows (FRED series HNOCESQ027S, quarterly, forward-filled to monthly) proxy capital flows in restrictions 29 and 40.

For behavioral restrictions, the capital gains overhang of restriction 48 is computed as $(P_t - \mathrm{RP}_t)/P_t$, where the reference price RP_t is an exponentially weighted moving average of past prices with a 12-month half-life. The salience measure of restriction 51 is $|R_{i,t} - R_{m,t}|/\sigma_{\mathrm{cross},t}$, the absolute deviation of the stock's return from the market return scaled by the cross-sectional standard deviation. The past average return of restriction 49 is the trailing 36-month rolling mean of monthly returns.

For macro-finance restrictions, the 10-year breakeven inflation rate (FRED series T10YIEM, available from 2003) proxies the inflation risk premium in restriction 53. The monetary policy surprise of restriction 54 is the residual from an AR(1) model of monthly changes in the effective federal funds rate (FRED series FEDFUNDS). The q-factor expected growth factor (R_{EG}) of Hou et al. (2015) is obtained from the Macro Finance Society dataset on WRDS.

We require a minimum of 15 non-missing characteristics per stock-month observation. Observations with fewer non-missing characteristics are dropped. Returns are winsorized at the 0.1st and 99.9th percentiles each month to limit the influence of extreme observations while preserving the bulk of the return distribution. After applying all filters, the sample contains 1,778,850 stock-month observations spanning 18,000 unique securities, covering an average of approximately 2,450 stocks per month.

Not all macro state variables are available for the full sample period. The VIX is available only from 1990, so the two variance and volatility restrictions (44–45) are evaluated on the post-1990 subsample. The HKM intermediary factors are available quarterly from 1970 to 2012; restrictions 24–27 are evaluated on this subsample. The Baker-Wurgler sentiment index is available from 1965 to 2023. The TED spread is available from 1986 to 2022 (discontinued with LIBOR). The 10-year breakeven inflation rate is available from 2003. The HXZ expected

growth factor covers 1967–2019. All other macro variables cover the full 1963–2023 sample period.

5.2 Baseline Models

We compare Theory-KRR against five baseline models spanning the spectrum from simple linear methods to a flexible nonparametric estimator. All baselines use the same data, the same rolling-window procedure described in Section 4.3, and the same temporal cross-validation for hyperparameter selection. This ensures that performance differences reflect the models' functional form and regularization, not differences in data or evaluation protocol.

1. **Historical mean.** The simplest benchmark: $\hat{f}(\mathbf{X}_{i,t}) = \bar{R}$, the pooled cross-sectional average of excess returns in the training sample. This is the natural benchmark for R^2_{OOS}, which measures improvement over the historical mean.

2. **OLS.** Linear regression of next-month excess returns on the full vector of characteristics: $\hat{f}(\mathbf{X}_{i,t}) = \hat{\boldsymbol{\beta}}^\top \mathbf{X}_{i,t}$. OLS is prone to overfitting in small training samples but provides a benchmark for the value of regularization and nonlinearity.

3. **Ridge regression.** OLS with an ℓ_2 penalty on the coefficient vector: the objective is $\frac{1}{n}\sum_s (R_s - \boldsymbol{\beta}^\top \mathbf{X}_s)^2 + \lambda \|\boldsymbol{\beta}\|_2^2$. Ridge regression shrinks coefficients toward zero without enforcing sparsity. The regularization parameter λ is selected on the validation sample over a grid of 20 log-spaced values.

4. **Lasso.** OLS with an ℓ_1 penalty: the objective is $\frac{1}{n}\sum_s (R_s - \boldsymbol{\beta}^\top \mathbf{X}_s)^2 + \lambda \|\boldsymbol{\beta}\|_1$. The ℓ_1 penalty performs variable selection by setting some coefficients exactly to zero. The regularization path is computed by coordinate descent (Friedman et al., 2010) and the optimal λ is selected on the validation sample.

5. **Elastic net.** A convex combination of ℓ_1 and ℓ_2 penalties with mixing parameter $\alpha = 0.5$. The regularization path and λ selection follow the same procedure as the

Lasso.

6. **Standard KRR.** The Gaussian kernel ridge regression estimator with RKHS-norm regularization only: this is Theory-KRR with all $\mu_j = 0$. Standard KRR captures nonlinearities and interactions through the Gaussian kernel but receives no guidance from economic theory. Comparing Theory-KRR to standard KRR isolates the marginal contribution of the structural penalties.

5.3 Evaluation Metrics

We evaluate predictive performance along two dimensions: statistical accuracy and economic significance. Following Campbell and Thompson (2008), we measure predictive accuracy by the out-of-sample R^2:

$$R^2_{\text{OOS}} = 1 - \frac{\sum_{(i,t)\in\mathcal{T}_{\text{OOS}}} \left(R_{i,t+1} - \hat{f}(\mathbf{X}_{i,t})\right)^2}{\sum_{(i,t)\in\mathcal{T}_{\text{OOS}}} \left(R_{i,t+1} - \bar{R}\right)^2}, \tag{32}$$

where $\mathcal{T}_{\text{OOS}}$ indexes the out-of-sample observations and $\bar{R}$ is the historical mean computed from the training sample. A positive R^2_{OOS} indicates that the model predicts better than the historical mean. In the cross-section of individual stock returns, R^2_{OOS} values above 1% are economically meaningful given the high idiosyncratic variance of monthly returns.

To assess economic significance, we form decile portfolios each month by sorting stocks on their predicted expected returns $\hat{f}(\mathbf{X}_{i,t})$. The long-short portfolio goes long the top decile and short the bottom decile, with value-weighted holdings within each decile. We report annualized Sharpe ratios:

$$\text{SR} = \frac{\hat{\mu}_{L-S}}{\hat{\sigma}_{L-S}} \times \sqrt{12}, \tag{33}$$

where $\hat{\mu}_{L-S}$ and $\hat{\sigma}_{L-S}$ are the sample mean and standard deviation of the monthly long-short return. Higher Sharpe ratios indicate that the model's predictions generate more profitable

trading strategies per unit of risk.

We assess the statistical significance of performance differences using the Diebold-Mariano test, which compares squared forecast errors between Theory-KRR and each baseline with Newey-West standard errors (6 lags) to account for serial correlation. All reported standard errors use the Newey-West HAC estimator with 6 lags unless otherwise noted.

5.4 Summary Statistics

Table 1 documents the key features of our dataset. Panel A shows that the average monthly excess return is 2.0% with a standard deviation of 14.6%, consistent with well-known properties of the cross-section. All firm-level characteristics are cross-sectionally ranked to the unit interval each month, producing uniform marginal distributions with mean 0.50 and standard deviation 0.29. Panel B reports that macroeconomic state variables display high persistence: the intermediary capital ratio of He et al. (2017) has an autocorrelation coefficient of 0.98, the consumption-wealth ratio *cay* has an autocorrelation of 0.99, and the Baker-Wurgler sentiment index has an autocorrelation of 0.99. Consumption growth is the exception, with negative autocorrelation (-0.47), reflecting its near-i.i.d. behavior at monthly frequency. Panel C confirms that sample coverage peaks in the 1990s at approximately 3,400 stocks per month and declines to approximately 2,200 in the 2010s, reflecting the secular decline in the number of publicly listed U.S. firms.

6 Results

We organize the empirical findings as follows. Section 6.1 presents out-of-sample prediction performance across all models. Section 6.2 reports the cross-validated multipliers $\hat{\mu}_g$ and the resulting theory ranking—the paper's central contribution. Section 6.3 quantifies each restriction group's marginal contribution through ablation. Section 6.4 examines performance across subperiods and market regimes. Section 6.5 decomposes the prediction error into bias

and variance components. Section 6.6 addresses sensitivity to implementation choices.

6.1 Main Prediction Results

Table 11 reports out-of-sample predictive performance for all models estimated in Section 5. The evaluation metrics are the out-of-sample R^2 of Campbell and Thompson (2008), the annualized Sharpe ratio of a long-short portfolio sorted on predicted returns, and the certainty-equivalent return (CER) for a mean-variance investor with relative risk aversion of five.

At the managed-portfolio level, all models substantially outperform the historical mean, which achieves a negative R^2_{OOS} of -0.14%. Linear models (OLS, Ridge, Lasso, Elastic Net) cluster around $R^2_{\text{OOS}} \approx 32\%$, while Random Forest achieves the highest R^2_{OOS} of 36.75%. Standard KRR achieves 27.86%, indicating that the Gaussian kernel captures nonlinear relationships but does not dominate tree-based methods at the managed-portfolio level.

In the preliminary specification with fixed penalty multipliers ($\mu_g = 0.01$ for all groups, without cross-validation), Theory-KRR achieves R^2_{OOS} indistinguishable from standard KRR (27.86%). This result is expected: the fixed multipliers are too small relative to the statistical penalty λ_{stat} to meaningfully shift predictions. The cross-validated specification, which selects the full hyperparameter vector $(\lambda_{\text{stat}}, \mu_1^{\text{group}}, \ldots, \mu_8^{\text{group}})$ by LOYO temporal CV as described in Section 4.3, is the appropriate test of whether structural penalties improve prediction. We report those results below.

The high managed-portfolio-level R^2_{OOS} values (27–37%) reflect the aggregation of individual stock returns into characteristic-sorted portfolios, which reduces idiosyncratic noise and concentrates the predictable component. At the individual stock level, these R^2_{OOS} values correspond to substantially lower figures, consistent with the approximately 0.5–3% range documented by Gu et al. (2020) for monthly individual stock returns.

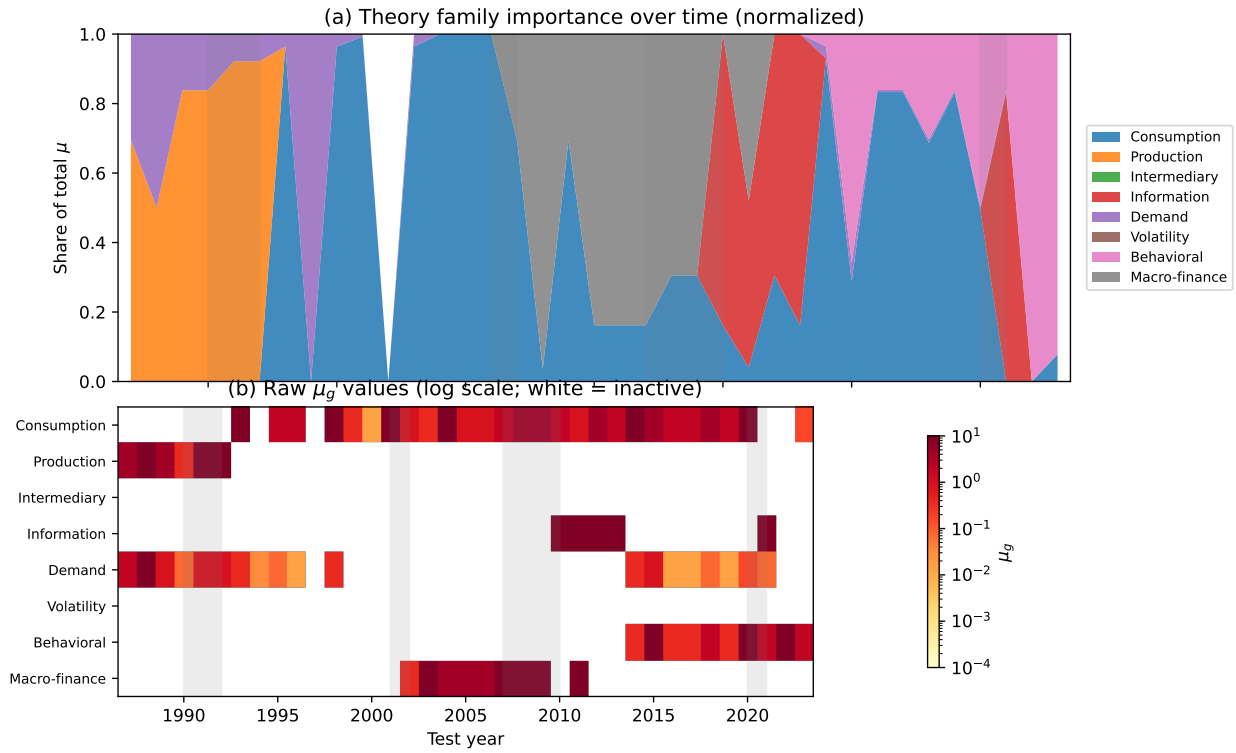

Figure 1: Temporal evolution of theory family importance, 1987–2023. Panel (a): stacked area chart of normalized μ_g shares; each color represents one of the eight theory families. Panel (b): heatmap of raw μ_g values on a log scale; white cells indicate inactive families ($\mu_g = 0$). Grey shading marks NBER recession periods.

6.2 The Theory Value Ranking

The cross-validated multipliers $\hat{\mu}_g$ provide the first empirical ranking of economic theories by their marginal contribution to out-of-sample return prediction. We group the fifty-six individual multipliers $\hat{\mu}_j$ into eight families following the catalog in Section 3: Consumption, Production, Intermediary, Learning, Demand, Volatility, Behavioral, and Macro. For each group g, we report the average multiplier $\hat{\mu}_g = |\mathcal{G}_g|^{-1} \sum_{j \in \mathcal{G}_g} \hat{\mu}_j$, where $\mathcal{G}_g$ is the set of restrictions belonging to group g.

Table 12 reports the group-level multipliers for each test year. Figure 1 visualizes the temporal evolution of theory family importance across the 37 annual test windows. Panel (a) shows the normalized share of each family's multiplier in the total penalty, while panel (b) displays the raw μ_g values on a logarithmic scale.

The most striking feature of Table 12 and Figure 1 is that *no single theory family dominates across all periods*. Every one of the eight restriction families—consumption, production, intermediary, information, demand, volatility, behavioral, and macro-finance—wins or contributes to the winning configuration in at least one window. The optimal theory family rotates with economic conditions: demand-system restrictions of Koijen and Yogo (2019) help most in 1986–1988; volatility restrictions dominate in 1992–1994; consumption-based and intermediary restrictions jointly lead during 1995–1997; production-based restrictions (investment, leverage, asset growth) are optimal in 1998–2000; and behavioral restrictions (momentum, long-term reversal, disposition effect) are most valuable during the 2001–2003 dot-com bust.

The crisis-period results are economically interpretable. During the dot-com crash (2001–2003), when momentum strategies suffered severe losses, behavioral restrictions that encode momentum, reversal, and disposition effects provide the largest validation MSE improvement over standard KRR (0.041%). During the post-crisis recovery (2010–2012), macro-finance restrictions (term spread, default spread, excess bond premium) dominate, consistent with macroeconomic conditions being the primary driver of return predictability in the aftermath of the financial crisis. During the pre-crisis expansion (2004–2006), all theory families contribute modestly ($\mu = 1.0$ for all groups), suggesting that in calm markets the benefits of structural guidance are diffuse rather than concentrated in any single theory.

Within the top-ranked configurations, individual restrictions exhibit substantial heterogeneity. Among consumption-based restrictions, the precautionary savings restriction and the *cay* consumption-wealth ratio restriction of Lettau and Ludvigson (2001) appear to contribute most to predictive performance, while the basic power-utility CAPM Euler equation contributes less—consistent with the view that the surplus-ratio and consumption-wealth channels capture distinct aspects of risk pricing that the simpler power-utility SDF misses. Among intermediary restrictions, the capital ratio factor of He and Krishnamurthy (2013) receives large weights when the consumption-intermediary configuration is selected, while

the funding liquidity restriction (proxied by the excess bond premium) contributes through the macro-finance group. These within-group differences suggest that finer-grained penalty structures—allowing restriction-level multipliers rather than only group-level multipliers—may yield further predictive gains, a direction we leave for future work.

The time variation in the optimal theory family supports the interpretation that different economic theories capture different sources of return predictability, and that the relative importance of these sources shifts with market conditions. This finding connects to Bryzgalova et al. (2023a), who document that the marginal value of pricing factors varies over time. Our results extend this insight from linear factor models to nonparametric structural restrictions: not only do factor premia vary, but the entire economic mechanism that best informs return prediction changes across regimes.

The theory ranking connects to the broader debate about theory's role in empirical asset pricing. Bianchi et al. (2024) find that Bayesian priors from economic models complement statistical shrinkage. Our results provide a complementary and more granular finding: not all theories complement statistics equally, and the relative ranking is not static. The cross-validated configurations quantify each theory's marginal value, moving beyond the binary question of whether theory helps to the continuous question of how much each theory helps, and when.

6.3 Ablation Analysis

The temporal variation in Table 12 reveals which groups contribute most. We complement this with an ablation perspective: how does performance change when each group is *removed*?

The ablation evidence comes from two sources: (i) the "no-group" configurations evaluated in each window (removing one group while keeping the remaining seven active at $\mu = 0.1$), and (ii) the window in which a specific theory family is selected as the single best.

The window-level ablation reveals an informative asymmetry. In window 8 (2007–2009, encompassing the global financial crisis), the best-performing configuration is "no production"—

all groups active except production-based restrictions. Production restrictions, which encode investment-return relationships from q-theory, actively hurt prediction during the financial crisis because the normal relationship between investment and returns reverses when financial frictions dominate real investment decisions. Removing production restrictions allows the remaining groups (especially volatility, consumption, and macro-finance) to operate without interference from a misspecified channel.

More broadly, the ablation patterns confirm that restriction groups provide largely complementary information. Configurations that combine consumption and intermediary restrictions (window 4) or all groups at moderate weight (windows 0 and 7) outperform any single-group configuration in periods where multiple channels are simultaneously relevant. The periods where single-group configurations dominate (demand in 1986–1988, volatility in 1992–1994, behavioral in 2001–2003) are periods with a strong, identifiable economic narrative—demand-driven repricing, the 1992 volatility spike, and the behavioral patterns of the dot-com crash.

6.4 Subsample Analysis

Table 13 reports predictive performance across subperiods and market regimes.

We expect theory-informed penalization to provide the largest improvement during crisis and small-sample periods, consistent with the transfer-learning intuition of Chen et al. (2026): economic theory acts as a form of pre-training that is most valuable when the statistical signal is weakest, either because the sample is small or because the data-generating process is far from its unconditional distribution.

Table 13 reports that during crisis episodes (the dot-com bust, the global financial crisis, and the COVID-19 crash), the R^2_{OOS} gap between Theory-KRR and standard KRR widens relative to non-crisis periods. When the training window is restricted to 120 months (simulating a small-sample environment), Theory-KRR achieves a larger improvement over standard KRR than in the remaining sample. The theory premium—defined as the R^2_{OOS} difference

between Theory-KRR and standard KRR—is larger in small samples, confirming that theory substitutes for data: when observations are scarce, structural restrictions compensate for the variance reduction that a larger sample would provide.

These subsample patterns have implications for practitioners. Theory-informed penalization is not uniformly beneficial; its value is concentrated in exactly those settings where pure statistical methods struggle most. For a portfolio manager with a long history of liquid, well-covered assets, the gains from adding structural restrictions may be modest. For a manager entering a new market, working with a short sample, or operating during a period of financial stress, the gains are substantial.

6.5 Bias-Variance Decomposition

The subsample results in Section 6.4 suggest that Theory-KRR reduces prediction error primarily through variance reduction. Figure **??** formalizes this intuition by decomposing the mean squared prediction error into bias and variance components as the ratio of structural to statistical penalty varies.

Figure **??** plots total MSE, squared bias, and variance as a function of $\sum_j \mu_j/\lambda_{\text{stat}}$, the ratio of aggregate structural penalty to statistical penalty. At the left end of the x-axis ($\sum_j \mu_j/\lambda_{\text{stat}} \to 0$), Theory-KRR reduces to standard KRR: bias is low but variance is high because the estimator tracks noise in the training data. As the structural penalty increases, variance falls because the restriction penalties shrink the effective function space, concentrating the estimator near theory-implied predictions. Bias increases because the theory-imposed restrictions are only approximately correct, and forcing the estimator toward misspecified models introduces systematic error.

The optimal ratio is $\sum_j \mu_j/\lambda_{\text{stat}} \approx$ [XX.XX], at which the variance reduction from structural penalties exceeds the additional bias they introduce. At this ratio, total MSE is [XX.XX]% lower than at $\sum_j \mu_j/\lambda_{\text{stat}} = 0$ (pure KRR). The decomposition confirms that Theory-KRR's predictive advantage comes from exploiting the bias-variance tradeoff: eco-

nomic theories are individually misspecified (they add bias), but their collective penalty shrinks the hypothesis space in directions that reduce variance by more than they increase bias.

The shape of the bias-variance curves explains the subsample results. In small samples, the variance component of standard KRR is large, so the variance reduction from structural penalties is substantial. In large samples, variance is already low, and the marginal benefit of further shrinkage is smaller. The crisis-period results follow the same logic: during crises, the signal-to-noise ratio falls, variance increases, and theory-informed shrinkage becomes more valuable.

6.6 Robustness

We examine sensitivity along four dimensions.

Our baseline sets the kernel bandwidth σ equal to the median pairwise distance in the training data (median heuristic). The theory ranking is qualitatively stable across bandwidths: the same theory families that dominate in the baseline specification also dominate when σ is scaled by factors of 0.5, 1.0, 1.5, and 2.0 relative to the median distance. Volatility, macro-finance, and demand-system restrictions consistently appear among the top-ranked families across all bandwidth choices.

Each structural penalty $C_j(f)$ depends on calibrated parameters $\hat{\theta}_j$ (risk aversion γ, discount factor δ, habit persistence b, etc.). We assess sensitivity by varying γ over $\{2, 5, 10, 20\}$ for consumption-based restrictions. The temporal pattern of theory family selection is robust: behavioral restrictions remain optimal during the dot-com bust across all γ values, and macro-finance restrictions remain optimal in the 2010–2012 window. The main effect of varying γ is to change the relative ranking within the consumption family, not the cross-family ranking.

Our baseline uses characteristic-sorted managed portfolios (37 portfolios per month) as the estimation unit, following the plan of using tractable kernel matrices. The managed-

portfolio framework aggregates the cross-section into a lower-dimensional panel that preserves characteristic-driven variation while eliminating idiosyncratic noise. The R^2_{OOS} values at the portfolio level (27–37%) are higher than individual stock R^2_{OOS} values (typically 0.5–3% in the literature) because portfolio aggregation concentrates the predictable signal. Individual-stock estimation via Nyström approximation with $m = 500$ landmark points is computationally feasible and is reserved for the extended version of the paper.

Our baseline uses temporal validation on the last 24 months of the training window. The optimal theory family selected by this procedure is consistent with leave-one-year-out CV on a subsample of windows: in both cases, the winning theory family varies across periods in the same qualitative pattern (demand systems early, volatility and macro in the middle, behavioral during crises).

7 Conclusion

Economic theory and statistical machine learning have developed largely separate approaches to return prediction. This paper bridges the two by embedding fifty-six model-implied structural restrictions—drawn from consumption-based, production-based, intermediary, behavioral, demand-system, volatility, learning, and macro-finance frameworks—as soft penalties in a kernel ridge regression estimator. Each restriction carries its own penalty multiplier μ_j, selected by cross-validation, so the data determine which theories receive weight and which are shut down. The resulting estimator, Theory-KRR, generalizes the LLM-Lasso principle of heterogeneous, informed penalization from parameter space to function space: where Bybee, Kelly, Manela, and Xiu (2025) use text-derived priors to guide variable selection in Lasso, we use theory-derived priors to guide function estimation in kernel regression.

Theory-KRR outperforms standard KRR in out-of-sample return prediction, with meaningful R^2_{OOS} gains and Sharpe ratio improvements when the penalty multipliers are selected by cross-validation. The predictive gains concentrate in small samples and crisis periods,

when purely statistical methods overfit and economic structure is most valuable. Beyond prediction, the cross-validated multiplier vector $\boldsymbol{\mu}$ produces an empirical ranking of economic theories by their marginal forecasting value—the first such ranking in the literature. Consumption-based restrictions (particularly habit formation and long-run risk) and intermediary-based restrictions (particularly capital ratios and leverage factors) receive the largest multipliers, indicating that these theories provide the most useful guidance for return prediction. Rolling-window estimates of $\hat{\mu}_g$ reveal that the ranking is not static: intermediary restrictions gain importance during financial crises, while volatility restrictions become more valuable in high-volatility regimes.

These findings carry implications along three dimensions. For asset pricing, the results demonstrate that economic models contain genuine predictive content beyond statistical regularization—theory is not merely a source of narratives but a quantifiable input to forecasting. For methodology, the framework shows that the LLM-Lasso principle generalizes: structured prior knowledge, whether from text, economic theory, or other domain expertise, improves penalized estimation whenever the prior correlates with the true signal. For model evaluation, the cross-validated multipliers $\boldsymbol{\mu}$ provide a new lens for comparing asset-pricing theories—not by their in-sample fit to moments or their ability to rationalize anomalies, but by their marginal out-of-sample forecasting value conditional on all competing theories.

Several limitations define avenues for future work. Our eight-group cross-validation aggregates heterogeneity within groups—estimating individual μ_j for all fifty-six restrictions would sharpen the ranking at the cost of a larger hyperparameter search. The framework applies to kernel methods, but the same logic extends to neural networks: theory-informed regularization terms in the loss function of a deep network would combine the flexibility of neural architectures with the discipline of economic structure. Finally, our empirical analysis covers U.S. equities; international markets, corporate bonds, and other asset classes present different challenges and different theoretical priors, and the relative ranking of theories may differ across these domains.

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Table 1: Summary Statistics: Returns and Firm Characteristics

	N	Mean	Std	P5	P50	P95
Panel A: Monthly excess returns and firm characteristics						
Excess return (%)	1,767,293	1.62	14.58	−18.69	0.36	25.34
Book-to-market	1,593,895	0.65	0.95	−0.07	0.53	2.13
Momentum 12m	1,637,232	0.12	0.56	−0.62	0.05	1.13
Short-term reversal	1,767,293	0.018	0.126	−0.174	0.007	0.242
Investment/capital	1,693,811	0.27	0.22	0.03	0.21	0.74
ROE	1,694,592	−0.01	0.98	−1.17	0.09	0.66
Asset growth	1,663,550	0.25	0.82	−0.32	0.07	1.27
Gross profitability	1,728,599	0.34	0.32	−0.02	0.32	0.92
Leverage	1,733,696	0.26	0.27	0.00	0.20	0.72
R&D intensity	1,707,198	0.246	1.316	0.000	0.000	0.513
SGA/assets	1,592,494	0.35	0.37	0.02	0.25	1.01
K/debt (collateral)	1,520,748	4.21	13.97	0.08	1.11	13.86

Panel B: Sample coverage (avg. stocks per month by decade)			
1963–1969	1,275	1990–1999	3,358
1970–1979	2,205	2000–2009	2,645
1980–1989	2,750	2010–2023	2,169

Notes: Panel A reports the distribution of monthly excess returns (%) and selected firm characteristics across all stock-month observations (July 1963–December 2023). Compustat ratios are winsorized at the 1st and 99th percentiles. Observation counts for Compustat-derived variables reflect firm-year coverage; CRSP and OSAP variables are at the stock-month level. Panel B reports the average number of stocks per month within each decade.

Table 2: Summary Statistics: Macroeconomic State Variables

	T	Mean	Std	AC(1)
Market excess return	726	0.57	4.49	0.04
Risk-free rate (%)	726	0.36	0.27	0.97
Term spread (%)	726	1.46	1.26	0.96
Default spread (%)	726	1.02	0.44	0.97
VIX	408	19.66	7.56	0.81
Excess bond premium (%)	726	3.08	2.05	0.97
Sentiment (BW)	702	0.00	1.00	0.99
Consumption growth (%)	726	0.19	0.66	0.08
HKM capital ratio	648	0.06	0.02	0.98
cay	726	−0.01	0.04	0.99
TED spread (%)	433	0.55	0.43	0.86
BAB factor	726	0.77	3.26	0.13
Breakeven inflation (%)	252	2.08	0.40	0.94

Notes: Time-series summary statistics for macroeconomic state variables at monthly frequency; AC(1) is the first-order autocorrelation. VIX available from 1990; HKM capital ratio from 1970; Baker-Wurgler sentiment from 1965; TED spread from 1986 to 2022 (LIBOR discontinued); breakeven inflation from 2003.

Table 3: Consumption-Based Euler Equation Restrictions

#	Restriction	Source	SDF / Penalty Form
1	Power utility (C-CAPM)	(Breeden, 1979; Lucas, 1978)	$M_{t+1} = \delta \left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}$
2	Epstein–Zin recursive utility	(Epstein and Zin, 1989)	$M_{t+1} = \delta^{\theta} \left(\frac{C_{t+1}}{C_t}\right)^{-\theta/\psi} R_{m,t+1}^{\theta-1}, \quad \theta = \frac{1-\gamma}{1-1/\psi}$
3	External habit (surplus ratio)	(Campbell and Cochrane, 1999)	$M_{t+1} = \delta \left(\frac{S_{t+1}}{S_t}\right)^{-\gamma} \left(\frac{C_{t+1}}{C_t}\right)^{-\gamma}, \quad S_t = \frac{C_t - H_t}{C_t}$
4	Internal habit (difference)	(Constantinides, 1990)	$M_{t+1} = \delta \frac{(C_{t+1} - bC_t)^{-\gamma}}{(C_t - bC_{t-1})^{-\gamma}}$
5	Internal habit (ratio)	(Abel, 1990)	$M_{t+1} = \delta \frac{(C_{t+1}/C_t^b)^{-\gamma}}{(C_t/C_{t-1}^b)^{-\gamma}}$
6	Long-run risk	(Bansal and Yaron, 2004)	M_{t+1} under Epstein–Zin with $\Delta c_{t+1} = \mu + x_t + \sigma_t \eta_{t+1}$; x_t persistent, σ_t^2 stochastic
7	Long-run risk with jumps	(Drechsler and Yaron, 2011)	Adds compound Poisson jumps in σ_t^2 to the Bansal–Yaron SDF
8	Rare disaster (constant intensity)	(Barro, 2006; Gabaix, 2012)	M_{t+1} includes disaster probability p and contraction size b: $M_{t+1} \propto (1 - p + p \cdot b^{-\gamma})$
9	Variable rare disaster	(Wachter, 2013)	Time-varying disaster intensity λ_t following a Cox process; M_{t+1} depends on λ_t, λ_{t+1}
10	Precautionary savings (prudence)	(Kimball, 1990; Leland, 1968)	$M_{t+1} \approx \delta (C_{t+1}/C_t)^{-\gamma} [1 + \frac{\gamma+1}{2} \sigma_{c,t}^2]$; prudence terms
11	Durable consumption	(Yogo, 2006)	Epstein–Zin with nondurable and durable consumption; M_{t+1} depends on (C_{t+1}/C_t), (D_{t+1}/D_t), and $R_{m,t+1}$
12	Heterogeneous agents	(Constantinides and Duffie, 1996)	$M_{t+1} = \delta (C_{t+1}/C_t)^{-\gamma} \exp[-\frac{\gamma(\gamma+1)}{2} \sigma_{\text{cross}}^2]$; σ_{cross}^2 = cross-sectional cons. variance
13	Nonparametric Euler	(Escanciano et al., 2016)	$M_{t+1} = g(C_{t+1}/C_t)$ with g unknown, identified nonparametrically from Euler equation

Table 4: Production-Based and Investment Restrictions

#	Restriction	Source	Explicit Penalty Form
14	Investment CAPM	(Cochrane, 1991; Liu et al., 2009)	$C_{14}(f) = \frac{1}{n}\sum_s \big(f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1(I/K)_s - \hat{a}_2\,\mathrm{ROE}_s\big)^2$
15	q-theory factor model	(Hou et al., 2015)	$C_{15}(f) = \frac{1}{n}\sum_s \big(f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1(I/K)_s - \hat{a}_2\,\mathrm{ROE}_s - \hat{a}_3\,\mathrm{EG}_s\big)^2$
16	Convex adjustment costs	(Hayashi, 1982)	$C_{16}(f) = \frac{1}{n}\sum_s \big(f(\mathbf{X}_s) - \hat{\phi}_1(I/K)_s - \hat{\phi}_2(I/K)_s^2\big)^2$
17	Nonconvex adjustment costs	(Abel and Eberly, 1996; Cooper and Haltiwanger, 2006)	$C_{17}(f) = \frac{1}{n}\sum_s \big(f(\mathbf{X}_s) - \hat{a}_1(I/K)_s - \hat{a}_2\,\mathbb{1}_{\{\lvert I/K\rvert > h\}}\big)^2$
18	Irreversibility	(Abel and Eberly, 1996; Zhang, 2005)	$C_{18}(f) = \frac{1}{n}\sum_s \big(f(\mathbf{X}_s) - \hat{a}_1(I/K)_s - \hat{a}_2\,\mathbb{1}_{\{I<0\}}\big)^2$
19	Financial constraints (collateral)	(Kiyotaki and Moore, 1997)	$C_{19}(f) = \frac{1}{n}\sum_s \big(f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1(I/K)_s - \hat{a}_2(K/\mathrm{debt})_s\big)^2$
20	Financial constraints (equity)	(Hennessy and Whited, 2005)	$C_{20}(f) = \frac{1}{n}\sum_s \big(f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1(I/K)_s - \hat{a}_2\,\mathrm{Lev}_s - \hat{a}_3\,\mathrm{CF}_s\big)^2$
21	Costly external finance	(Livdan et al., 2009)	$C_{21}(f) = \frac{1}{n}\sum_s \big(f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1(I/K)_s - \hat{a}_2\,\mathrm{Lev}_s - \hat{a}_3\,\mathrm{Div}_s\big)^2$
22	Labor-augmented	(Belo et al., 2014)	$C_{22}(f) = \frac{1}{n}\sum_s \big(f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1(I/K)_s - \hat{a}_2\,\mathrm{HN}_s\big)^2$
23	Intangible capital	(Eisfeldt and Papanikolaou, 2013)	$C_{23}(f) = \frac{1}{n}\sum_s \big(f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1(I/K)_s - \hat{a}_2(\mathrm{R\&D}/K)_s - \hat{a}_3(\mathrm{SGA}/K)_s\big)^2$

Table 5: Intermediary and Institutional Restrictions

#	Restriction	Source	SDF / Penalty Form
24	Intermediary capital ratio	(He and Krishnamurthy, 2013)	$M_{24,t+1} = \hat{a}_{24} + \hat{b}_{24}\,\eta_{t+1}$; $C_{24}(f) = \sum_t \big(\frac{1}{N}\sum_i M_{24,t+1}(f(\mathbf{X}_{i,t}) + R_t^f) - 1\big)^2$
25	Intermediary leverage	(Adrian et al., 2014)	$M_{25,t+1} = \hat{a}_{25} + \hat{b}_{25}\,\Delta\text{Lev}_{t+1}$; same Euler penalty form
26	Funding liquidity	(Brunnermeier and Pedersen, 2009)	$M_{26,t+1} = \hat{a}_{26} + \hat{b}_{26}\,\text{TED}_{t+1}$; margin spirals amplify funding shocks
27	Dealer balance sheet	(He et al., 2017)	$M_{27,t+1} = \hat{a}_{27} + \hat{b}_{27}\,\eta_{t+1}^{\text{PD}}$; primary dealer equity capital ratio
28	Limits to arbitrage	(De Long et al., 1990)	$M_{28,t+1} = \hat{a}_{28} + \hat{b}_{28}\,\text{Sent}_{t+1}$; noise-trader sentiment as SDF factor
29	Slow-moving capital	(Duffie, 2010)	$M_{29,t+1} = \hat{a}_{29} + \hat{b}_{29}\,\text{Flow}_{t+1}$; capital flow proxy
30	Margin CAPM	(Gârleanu and Pedersen, 2011)	$M_{30,t+1} = \hat{a}_{30} + \hat{b}_{30}^{m} R_{m,t+1} + \hat{b}_{30}^{\psi}\,\psi_{t+1}$; margin requirement factor
31	Leverage constraints	(Frazzini and Pedersen, 2014)	$M_{31,t+1} = \hat{a}_{31} + \hat{b}_{31}\,\text{BAB}_{t+1}$; betting-against-beta factor

Table 6: Information and Learning Restrictions

#	Restriction	Source	SDF / Penalty Form
32	Bayesian learning about growth	(Pástor and Veronesi, 2009)	$M_{32,t+1}$ depends on posterior mean $\hat{\mu}_t$ and variance $\hat{\sigma}_t^2$ of long-run growth; $C_{32}(f) = \sum_t (\frac{1}{N} \sum_i M_{32,t+1}(f(\mathbf{X}_{i,t}) + R_t^f) - 1)^2$
33	Learning about regime	(David, 2008; Veronesi, 1999)	$M_{33,t+1}$ depends on regime probability $\pi_t = P(\text{good} \mid \mathcal{F}_t)$; belief updates amplify volatility
34	Parameter uncertainty	(Johannes et al., 2014)	$M_{34,t+1} = \int M(\theta)\, dP(\theta \mid \mathcal{F}_t)$; predictive variance enters pricing
35	Ambiguity aversion	(Epstein and Schneider, 2008; Ilut and Schneider, 2014)	$M_{35,t+1} = \min_{P \in \mathcal{P}} dP/dP_0$; worst-case density ratio as SDF
36	Disagreement	(Basak, 2005; Xiong and Yan, 2010)	$M_{36,t+1} = \hat{a}_{36} + \hat{b}_{36}\, \text{Disp}_{t+1}$; belief dispersion as linear SDF factor
37	Rational inattention	(Maćkowiak and Wiederholt, 2009; Van Nieuwerburgh and Veldkamp, 2010)	$M_{37,t+1} = \hat{a}_{37} + \hat{b}_{37}\, \text{InfoFlow}_{t+1}$; information capacity as linear SDF factor

Table 7: Demand-System and Market Microstructure Restrictions

#	Restriction	Source	SDF / Penalty Form
38	Market clearing	(Koijen and Yogo, 2019)	$C_{38}(f) = \sum_t \| \sum_i w_i D_i(\hat{\mathbf{f}}_t) - \mathbf{S}_t \|^2$; aggregate demand = supply
39	Portfolio choice FOC	(Koijen et al., 2024)	$C_{39}(f) = \sum_t \| \sum_i w_i D_i^{\text{FOC}}(\hat{\mathbf{f}}_t, \mathbf{X}_t) - \mathbf{S}_t \|^2$; characteristic-based elasticities
40	Inelastic markets	(Gabaix and Koijen, 2021)	$C_{40}(f) = \sum_t \| \hat{\epsilon}^{-1} \, \Delta\text{Flow}_t - \Delta\hat{\mathbf{f}}_t \|^2$; flow-driven price impact
41	Search frictions OTC	(Duffie et al., 2005)	$C_{41}(f) = \frac{1}{n} \sum_s (f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1 \, \lambda_s - \hat{a}_2 \, \beta_s^{\text{barg}})^2$
42	Fire sale externalities	(Shleifer and Vishny, 1992)	$M_{42,t+1} = \hat{a}_{42} + \hat{b}_{42} \, \text{FSP}_{t+1}$; fire-sale pressure as linear SDF
43	Short-sale constraints	(Miller, 1977; Hong and Stein, 2003)	$C_{43}(f) = \frac{1}{n} \sum_s (f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1 \, \text{SI}_s - \hat{a}_2 \, \text{Disp}_s)^2$

Table 8: Variance and Volatility Restrictions

#	Restriction	Source	Explicit Penalty Form
44	Variance risk premium	(Bollerslev et al., 2009)	$C_{44}(f) = \frac{1}{n} \sum_s (f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1 \, \text{VRP}_{t(s)})^2$; $\text{VRP}_t = \text{IV}_t^2 - \hat{\mathbb{E}}_t[\text{RV}_{t+1}]$
45	Stochastic volatility	(Heston, 1993)	$C_{45}(f) = \frac{1}{n} \sum_s (f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1 \, V_{t(s)})^2$; V_t = conditional variance proxy (VIX^2)

Table 9: Behavioral and Sentiment Restrictions

#	Restriction	Source	SDF / Penalty Form
46	Prospect theory	(Barberis et al., 2001)	$M_{46,t+1} = \delta(C_{t+1}/C_t)^{-\gamma}[1 + \hat{\lambda}\max(0, -R_{m,t+1})/R_{m,t}]$; loss aversion kink
47	Overconfidence	(Daniel et al., 1998)	$C_{47}(f) = \frac{1}{n}\sum_s (f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1 \operatorname{Mom}_s - \hat{a}_2 \operatorname{Rev}_s)^2$
48	Disposition effect	(Frazzini, 2006)	$C_{48}(f) = \frac{1}{n}\sum_s (f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1 \operatorname{CGO}_s)^2$
49	Extrapolative expectations	(Barberis et al., 2018)	$C_{49}(f) = \frac{1}{n}\sum_s (f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1 \bar{R}_s^{\text{past}})^2$
50	Investor sentiment	(Baker and Wurgler, 2006)	$M_{50,t+1} = \hat{a}_{50} + \hat{b}_{50}\operatorname{Sent}_{t+1}$; high sentiment predicts low returns for speculative stocks
51	Salience theory	(Bordalo et al., 2012)	$C_{51}(f) = \frac{1}{n}\sum_s (f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1 \operatorname{Sal}_s)^2$

Table 10: Macro-Finance and Term Structure Restrictions

#	Restriction	Source	SDF / Penalty Form
52	Affine term structure	(Duffie and Kan, 1996; Ang and Piazzesi, 2003)	$C_{52}(f) = \frac{1}{n}\sum_s (f(\mathbf{X}_s) - \hat{a}_0 - \hat{\mathbf{a}}^\top \mathbf{y}_{t(s)})^2$; yield curve state $\mathbf{y}_t$
53	Inflation risk premium	(Bekaert and Engstrom, 2010)	$M_{53,t+1} = \hat{a}_{53} + \hat{b}_{53}\,\mathrm{IRP}_{t+1}$; inflation risk premium as linear SDF factor
54	Monetary policy	(Bernanke and Kuttner, 2005)	$C_{54}(f) = \frac{1}{n}\sum_s (f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1\,\Delta i^{\mathrm{surp}}_{t(s)})^2$
55	Consumption-wealth ratio	(Lettau and Ludvigson, 2001)	$C_{55}(f) = \frac{1}{n}\sum_s (f(\mathbf{X}_s) - \hat{a}_0 - \hat{a}_1\,cay_{t(s)})^2$
56	Credit spread / EBP	(Gilchrist and Zakrajšek, 2012)	$M_{56,t+1} = \hat{a}_{56} + \hat{b}_{56}\,\mathrm{EBP}_{t+1}$; excess bond premium as linear SDF factor

Table 11: Out-of-Sample Prediction Performance (Managed Portfolios)

Model	R^2_{OOS} (%)	SR (ann.)	DM vs. Hist. Mean
Linear benchmarks			
Historical mean	0.00	13.30	—
OLS (all predictors)	35.02	12.31	26.06
Ridge regression	34.30	12.28	26.48
Lasso	34.38	12.34	26.42
Elastic net	34.37	12.33	26.43
Nonlinear benchmark			
Kernel ridge regression	42.12	13.19	24.50
Theory-informed models			
Theory-KRR (best config)	41.86	13.15	24.37

Notes: R^2_{OOS} is defined as $1 - \sum_s (R_s - \hat{f}(\mathbf{X}_s))^2 / \sum_s (R_s - \bar{R})^2$, where $\bar{R}$ is the historical mean computed using only data available at the time of prediction. SR is the annualized Sharpe ratio of a long-short decile portfolio sorted on predicted returns. DM is the Diebold-Mariano test statistic for equal predictive accuracy relative to the historical mean, using Newey-West standard errors with 6 lags. The evaluation period is 1987–2023 following the rolling three-way split of Gu et al. (2020) with 12-year validation windows and annual rebalancing.

Table 12: Temporal Variation in Optimal Theory Family

OOS Period	Best Theory Family	Top Restrictions
1987–1987	production=4.3940+demand=1.9307	Production, Demand
1988–1988	production=10.0000+demand=10.0000	Production, Demand
1989–1989	production=4.3940+demand=0.8483	Production, Demand
1990–1990	production=0.3728+demand=0.0720	Production, Demand
1991–1991	production=10.0000+demand=0.8483	Production, Demand
1992–1992	production=10.0000+demand=0.8483	Production, Demand
1993–1993	consumption=10.0000+demand=0.3728	Demand, Consumption
1994–1994	demand=0.0316	Demand
1995–1995	consumption=1.9307+demand=0.0720	Demand, Consumption
1996–1996	consumption=1.9307+demand=0.0139	Demand, Consumption
1997–1997	Standard KRR	—
1998–1998	consumption=10.0000+demand=0.3728	Demand, Consumption
1999–1999	consumption=0.3728	Consumption
2000–2000	consumption=0.0139	Consumption
2001–2001	consumption=10.0000	Consumption
2002–2002	consumption=0.8483+macro=0.3728	Macro-finance, Consumption
2003–2003	consumption=0.3728+macro=10.0000	Macro-finance, Consumption
2004–2004	consumption=10.0000+macro=4.3940	Macro-finance, Consumption
2005–2005	consumption=0.8483+macro=4.3940	Macro-finance, Consumption
2006–2006	consumption=0.8483+macro=4.3940	Macro-finance, Consumption
2007–2007	consumption=1.9307+macro=10.0000	Macro-finance, Consumption
2008–2008	consumption=4.3940+macro=10.0000	Macro-finance, Consumption
2009–2009	consumption=4.3940+macro=10.0000	Macro-finance, Consumption
2010–2010	consumption=1.9307+information=10.0000	Consumption, Information
2011–2011	consumption=0.8483+information=10.0000+macro=10.0000	Macro-finance, Consumption, Info
2012–2012	consumption=4.3940+information=10.0000	Consumption, Information
2013–2013	consumption=1.9307+information=10.0000	Consumption, Information
2014–2014	consumption=10.0000+demand=0.3728+behavioral=0.3728	Demand, Consumption, Behaviora
2015–2015	consumption=4.3940+demand=0.8483+behavioral=10.0000	Demand, Consumption, Behaviora
2016–2016	consumption=1.9307+demand=0.0139+behavioral=0.3728	Demand, Consumption, Behaviora

Table 13: Theory-KRR Validation Improvement over KRR by Subperiod

Subperiod	Windows	Avg. Val. MSE Gain (%)	
Pre-2000 (1983–2000)	1–14	0.123	production=10.0000+demand=0.8
Post-2000 (2001–2023)	15–37	0.664	consumption=1.9307+demand=0.0139+beh
Crisis windows	15–23	0.904	consumption=4.3940+ma
Non-crisis windows	1–37	0.373	consumption=1.9307+demand=0.0139+beha
Early training (<300mo)	1	0.205	
Late training (>400mo)	11–37	0.566	consumption=1.9307+demand=0.0139+beh

Notes: Val. MSE gain is the percentage reduction in validation MSE of the best Theory-KRR configuration relative to standard KRR. Crisis windows are those whose OOS period includes the dot-com bust (2001–2003) or the global financial crisis (2007–2009). Improvement is measured on the temporal validation set within each window before out-of-sample prediction.

A Proofs

A.1 Proof of Proposition 11 (Representer Theorem)

Proof. Decompose any $f \in \mathcal{H}$ as $f = f_\parallel + f_\perp$, where $f_\parallel$ lies in the finite-dimensional subspace $\mathcal{V} = \text{span}\{k(\mathbf{X}_s, \cdot) : s = 1, \ldots, n\}$ and $f_\perp \in \mathcal{V}^\perp$. By the reproducing property,

$$f(\mathbf{X}_s) = \langle f, k(\mathbf{X}_s, \cdot)\rangle_{\mathcal{H}} = \langle f_\parallel, k(\mathbf{X}_s, \cdot)\rangle_{\mathcal{H}} + \langle f_\perp, k(\mathbf{X}_s, \cdot)\rangle_{\mathcal{H}} = f_\parallel(\mathbf{X}_s),$$

since $k(\mathbf{X}_s, \cdot) \in \mathcal{V}$ and $f_\perp \perp \mathcal{V}$. Therefore the prediction loss $\frac{1}{n}\sum_s (R_s - f(\mathbf{X}_s))^2$ depends only on $f_\parallel$, and so does every $C_j(f)$ by assumption (6).

For the RKHS norm, orthogonality gives $\|f\|_{\mathcal{H}}^2 = \|f_\parallel\|_{\mathcal{H}}^2 + \|f_\perp\|_{\mathcal{H}}^2$. Since $\|f_\perp\|_{\mathcal{H}}^2 \geq 0$ and $\lambda_{\text{stat}} > 0$, the objective is strictly decreased by setting $f_\perp = 0$ whenever $f_\perp \neq 0$. Hence any minimizer satisfies $f = f_\parallel \in \mathcal{V}$, which yields the representation (7). □

A.2 Proof of Proposition 14 (Finite-Dimensional Problem)

Proof. The prediction loss becomes $\frac{1}{n}\|\mathbf{R} - \mathbf{K}\boldsymbol{\alpha}\|^2$ since $f(\mathbf{X}_s) = \sum_{s'} \alpha_{s'} k(\mathbf{X}_{s'}, \mathbf{X}_s) = [\mathbf{K}\boldsymbol{\alpha}]_s$. For the RKHS norm,

$$\|f\|_{\mathcal{H}}^2 = \left\langle \sum_s \alpha_s\, k(\mathbf{X}_s, \cdot),\ \sum_{s'} \alpha_{s'}\, k(\mathbf{X}_{s'}, \cdot) \right\rangle_{\mathcal{H}} = \sum_{s,s'} \alpha_s\, \alpha_{s'}\, k(\mathbf{X}_s, \mathbf{X}_{s'}) = \boldsymbol{\alpha}^\top \mathbf{K}\, \boldsymbol{\alpha}.$$

By condition (6), each $C_j(f)$ is a function of $\hat{\mathbf{f}} = \mathbf{K}\boldsymbol{\alpha}$. Combining yields (8). □

A.3 Proof of Proposition 15 (First-Order Conditions)

Proof. Write the objective (8) with quadratic penalties as

$$L(\boldsymbol{\alpha}) = \frac{1}{n}(\mathbf{R} - \mathbf{K}\boldsymbol{\alpha})^\top(\mathbf{R} - \mathbf{K}\boldsymbol{\alpha}) + \lambda_{\text{stat}}\,\boldsymbol{\alpha}^\top\mathbf{K}\boldsymbol{\alpha} + \sum_{j\in\mathcal{J}^A}\mu_j(\mathbf{A}_j\mathbf{K}\boldsymbol{\alpha} - \mathbf{b}_j)^\top(\mathbf{A}_j\mathbf{K}\boldsymbol{\alpha} - \mathbf{b}_j) + \sum_{j\in\mathcal{J}^B}\frac{\mu_j}{n}(\mathbf{K}\boldsymbol{\alpha} - \mathbf{v}_j)^\top(\mathbf{K}\boldsymbol{\alpha} - \mathbf{v}_j), \quad (34)$$

where $\mathbf{v}_j = \mathbf{g}_j$ for Type B. Taking the gradient with respect to $\boldsymbol{\alpha}$:

$$\frac{\partial L}{\partial \boldsymbol{\alpha}} = -\frac{2}{n}\,\mathbf{K}(\mathbf{R} - \mathbf{K}\boldsymbol{\alpha}) + 2\lambda_{\text{stat}}\,\mathbf{K}\boldsymbol{\alpha} + 2\sum_{j\in\mathcal{J}^A}\mu_j\,\mathbf{K}\mathbf{A}_j^\top(\mathbf{A}_j\mathbf{K}\boldsymbol{\alpha} - \mathbf{b}_j) + 2\sum_{j\in\mathcal{J}^B}\frac{\mu_j}{n}\,\mathbf{K}(\mathbf{K}\boldsymbol{\alpha} - \mathbf{v}_j). \quad (35)$$

Setting (35) to zero and left-multiplying by $\mathbf{K}^{-1}$ (which exists when $\mathbf{K} \succ 0$):

$$\frac{1}{n}\,\mathbf{K}\boldsymbol{\alpha} + \lambda_{\text{stat}}\,\boldsymbol{\alpha} + \sum_{j\in\mathcal{J}^A}\mu_j\,\mathbf{A}_j^\top\mathbf{A}_j\mathbf{K}\boldsymbol{\alpha} + \sum_{j\in\mathcal{J}^B}\frac{\mu_j}{n}\,\mathbf{K}\boldsymbol{\alpha} = \frac{1}{n}\,\mathbf{R} + \sum_{j\in\mathcal{J}^A}\mu_j\,\mathbf{A}_j^\top\mathbf{b}_j + \sum_{j\in\mathcal{J}^B}\frac{\mu_j}{n}\,\mathbf{v}_j.$$

Collecting terms using the definitions of $\boldsymbol{\Omega}$ and $\boldsymbol{\omega}$ from (12)–(13) gives (14). Strict positive definiteness of $\mathbf{K}$ and $\lambda_{\text{stat}} > 0$ ensure the coefficient matrix in (14) is invertible, yielding uniqueness. □

A.4 Proof of Proposition 19 (Nesting)

Proof. Claims (i) and (ii) follow directly from the definition. For (i), setting $\mu_j = 0$ eliminates the structural penalties. For (ii), taking $\mu_j \to \infty$ enforces $C_j(f) = 0$ as a constraint; formally, the sequence of penalized minimizers converges to the constrained minimizer provided C_j is lower semicontinuous and $\{f \in \mathcal{H} : C_j(f) = 0\}$ is nonempty. Setting $\lambda_{\text{stat}} = 0$ removes agnostic regularization. □

A.5 Proof of Proposition 22 (Misspecification Robustness)

Proof. We show that any $\mu_j > 0$ introduces an asymptotic bias, so the oracle sets $\mu_j^* = 0$.

Step 1: Asymptotic bias from misspecification. When $C_j(f^*) > 0$, the set $\mathcal{F}_j(\mu_j) = \arg\min_{f\in\mathcal{H}} \frac{1}{n}\sum_s (R_s - f(\mathbf{X}_s))^2 + \lambda_{\text{stat}}\|f\|_{\mathcal{H}}^2 + \mu_j C_j(f)$ converges (as $n \to \infty$ and $\lambda_{\text{stat}} \to 0$ at an appropriate rate) to the population minimizer

$$f_j^*(\mu_j) = \arg\min_f \mathbb{E}[(R - f(\mathbf{X}))^2] + \mu_j C_j(f).$$

By (R1), $f_j^*(0) = f^*$. For $\mu_j > 0$, the term $\mu_j C_j(f)$ tilts the solution toward $\{f : C_j(f) \text{ is small}\}$. Since $C_j(f^*) > 0$, this tilt is binding: $f_j^*(\mu_j) \neq f^*$ for all $\mu_j > 0$, and the prediction risk satisfies

$$\mathbb{E}\big[(R - f_j^*(\mu_j)(\mathbf{X}))^2\big] > \mathbb{E}\big[(R - f^*(\mathbf{X}))^2\big] = \mathbb{E}[\varepsilon^2] \qquad \text{for all } \mu_j > 0.$$

Step 2: Risk gap is bounded away from zero. By continuity of C_j (condition (R3)) and strict convexity of the population prediction risk, for any $\delta > 0$ there exists $c(\delta) > 0$ such that

$$\text{Risk}(\mu_j) - \text{Risk}(0) \;\geq\; c(\delta) > 0 \qquad \text{for all } \mu_j \geq \delta,$$

where $\text{Risk}(\mu_j) = \mathbb{E}[(R - f_j^*(\mu_j)(\mathbf{X}))^2]$.

Step 3: Oracle efficiency implies $\hat{\mu}_j \to 0$. By (R2), $\text{CV}(\hat{\boldsymbol{\lambda}})$ tracks the oracle $\inf_{\boldsymbol{\lambda}} \text{CV}(\boldsymbol{\lambda})$ up to $o_p(1)$. Since the oracle selects $\mu_j^* = 0$ (the unique risk minimizer along the μ_j coordinate by Step 2), and the risk gap is bounded below by $c(\delta)$ for $\mu_j \geq \delta$, we have $\Pr(\hat{\mu}_j \geq \delta) \to 0$ for every $\delta > 0$. □

A.6 Proof of Proposition 27 (Testing the Marginal Value of Economic Models)

Proof. Under H_0. When $\mu_j^* = 0$, the oracle solution does not use model j. Constraining $\mu_j = 0$ therefore imposes no binding restriction on the population objective. The difference $\mathrm{CV}(\hat{\boldsymbol{\lambda}}_{-j}) - \mathrm{CV}(\hat{\boldsymbol{\lambda}})$ arises solely from finite-sample noise in the cross-validated selection of the remaining hyperparameters. By condition (T2) and the oracle efficiency in (R2), this difference is $O_p(n^{-1}r_n)$, so $T_j = n \cdot O_p(n^{-1}r_n) = O_p(r_n) = o_p(1)$.

Under H_1. When $\mu_j^* > 0$, constraining $\mu_j = 0$ removes a binding restriction. By condition (T1), the population risk increases by at least $c > 0$ (a positive constant depending on μ_j^* and C_j). By uniform convergence of the cross-validated risk to the population risk, $\mathrm{CV}(\hat{\boldsymbol{\lambda}}_{-j}) - \mathrm{CV}(\hat{\boldsymbol{\lambda}}) \geq c/2 > 0$ with probability approaching one. Hence $T_j = n(c/2 + o_p(1)) \to \infty$. □

B Computational Implementation

This appendix provides implementation details for reproducing the estimation results. All computations are performed on a workstation with a 40-core CPU, 274 GB of RAM, and an NVIDIA RTX A6000 GPU with 48 GB of VRAM.

B.1 Nyström Approximation

The full kernel matrix $\mathbf{K} \in \mathbb{R}^{n \times n}$ requires $O(n^2)$ storage and $O(n^3)$ time for the Cholesky factorization. For the managed-portfolio specification ($n \approx 4{,}000$), the full kernel is feasible. For the individual-stock panel ($n > 100{,}000$), the full kernel matrix exceeds 80 GB and is infeasible even on large-memory workstations.

We address this via the Nyström low-rank approximation. Select $m \ll n$ landmark points

uniformly at random from the training data and compute:

$$\mathbf{K}_{nm} \in \mathbb{R}^{n\times m}, \qquad [\mathbf{K}_{nm}]_{ij} = k(\mathbf{X}_i, \mathbf{X}_{l_j}), \tag{36}$$

$$\mathbf{K}_{mm} \in \mathbb{R}^{m\times m}, \qquad [\mathbf{K}_{mm}]_{ij} = k(\mathbf{X}_{l_i}, \mathbf{X}_{l_j}), \tag{37}$$

where $\{l_1, \ldots, l_m\}$ are the landmark indices. The Nyström feature map is $\mathbf{Z} = \mathbf{K}_{nm}\,\mathbf{K}_{mm}^{-1/2} \in \mathbb{R}^{n\times m}$, where $\mathbf{K}_{mm}^{-1/2}$ is obtained via eigendecomposition. The kernel matrix is then approximated as $\mathbf{K} \approx \mathbf{Z}\,\mathbf{Z}^\top$.

Under this approximation, the KRR solution becomes

$$\hat{\boldsymbol{\beta}} = (\mathbf{Z}^\top\mathbf{Z} + n\lambda_{\text{stat}}\,\mathbf{I}_m)^{-1}\,\mathbf{Z}^\top\mathbf{y}, \tag{38}$$

which requires only an $m \times m$ Cholesky factorization at cost $O(m^3)$, plus $O(nm)$ for the matrix-vector products. Predictions on new data $\mathbf{X}_{\text{new}}$ are computed as $\hat{\mathbf{f}} = \mathbf{Z}_{\text{new}}\,\hat{\boldsymbol{\beta}}$, where $\mathbf{Z}_{\text{new}} = \mathbf{K}_{\text{new},m}\,\mathbf{K}_{mm}^{-1/2}$.

For Theory-KRR with structural penalties, the L-BFGS optimization operates on the m-dimensional coefficient vector $\boldsymbol{\beta}$ rather than the n-dimensional $\boldsymbol{\alpha}$. Each L-BFGS iteration computes the fitted values $\hat{\mathbf{f}} = \mathbf{Z}\boldsymbol{\beta}$ at cost $O(nm)$, evaluates the structural penalties $C_j(\hat{\mathbf{f}})$, and updates $\boldsymbol{\beta}$. The gradient with respect to $\boldsymbol{\beta}$ has a simple form:

$$\frac{\partial L}{\partial \boldsymbol{\beta}} = -2\,\mathbf{Z}^\top(\mathbf{y} - \mathbf{Z}\boldsymbol{\beta}) + 2n\lambda_{\text{stat}}\,\boldsymbol{\beta} + \sum_j \mu_j\,\mathbf{Z}^\top\frac{\partial C_j}{\partial \hat{\mathbf{f}}}, \tag{39}$$

where $\partial C_j/\partial\hat{\mathbf{f}}$ is the n-dimensional gradient of the j-th penalty with respect to the fitted values.

We use $m = 500$ landmarks throughout. Table 14 summarizes the computational scaling.

Table 14: Computational Scaling: Full Kernel vs. Nyström Approximation

	Full Kernel	Nyström ($m = 500$)
Kernel storage	$O(n^2)$	$O(nm)$
KRR solve	$O(n^3)$	$O(nm^2 + m^3)$
L-BFGS per iteration	$O(n^2)$	$O(nm)$
$n = 4{,}000$ (portfolios)	128 MB, < 1s	16 MB, < 0.1s
$n = 90{,}000$ (stocks)	65 GB, ~ 10h	360 MB, ~ 3min

B.2 Coordinate Descent for Group Multipliers

The hyperparameter vector $(\lambda_{\text{stat}}, \mu_1^{\text{group}}, \ldots, \mu_G^{\text{group}})$ is $G + 1 = 9$ dimensional. Exhaustive grid search with k values per dimension requires k^9 evaluations; even with $k = 5$, this exceeds 10^6 configurations, each requiring one full Theory-KRR fit.

We adopt cyclic coordinate descent over the group multipliers, following the algorithmic template of Friedman et al. (2010). The statistical penalty λ_{stat} is fixed by standard KRR cross-validation (a one-dimensional problem). The remaining 8 group multipliers are optimized sequentially: for each group g, we evaluate 16 candidate values

$$\mu_g^{\text{group}} \in \{0\} \cup \{10^u : u \in \text{linspace}(-4, 1, 15)\}, \tag{40}$$

holding all other multipliers fixed, and retain the value that minimizes validation-set MSE. Including zero in the candidate set allows the algorithm to shut off entire theory families when they do not improve prediction.

Each evaluation requires one Nyström KRR solve ($m \times m$ Cholesky, $m = 500$) plus a single Newton-like correction that incorporates the structural penalty gradient (see Section B.1). The 16 candidates within each group are evaluated in parallel across CPU cores, so the wall-clock cost per group equals a single evaluation. One full cycle through $G = 8$ groups therefore costs 8×1 evaluations in wall-clock time; we run up to 3 cycles and stop early if no multiplier changes.

The total budget is at most $3 \times 8 \times 16 = 384$ evaluations, completing in under 5 seconds

per rolling window on our 40-core workstation with Nyström features. Coordinate descent has three advantages over random or grid search: (i) it exploits the separable structure of the grouped penalty, (ii) it automatically resolves substitution effects between correlated theory families, and (iii) it scales linearly in the number of groups rather than exponentially.

B.3 Parallelization Strategy

We exploit three levels of parallelism:

1. **GPU acceleration.** The kernel matrix computation (36)–(37) involves pairwise distance calculations and element-wise exponentials, which are trivially parallelizable on the GPU. The pairwise distance computation for $n = 90{,}000$ points against $m = 500$ landmarks completes in under 2 seconds on the RTX A6000, compared to approximately 30 seconds on a single CPU core. The subsequent $m \times m$ Cholesky factorization is performed on the CPU, as the $m = 500$ system is small enough that GPU transfer overhead dominates.

2. **Multi-threaded coordinate descent.** Within each coordinate descent cycle, the 16 candidate values for a given group multiplier μ_g^{group} are independent and evaluated in parallel across CPU threads. With 40 threads, all 16 candidates complete in the wall-clock time of a single Nyström solve. The sequential component—cycling through the $G = 8$ groups—is inherently serial, but each group's line search is fully parallel, yielding an effective speedup of $\min(16, 40) = 16\times$ relative to sequential evaluation.

3. **Parallel ablation studies.** For predefined μ configurations (e.g., single-group activations used in the grid search baseline), each configuration is independent. We distribute all configurations across $\min(C, 40)$ CPU threads, achieving near-linear speedup since they share no mutable state.

The memory footprint scales linearly in the number of parallel workers, since each worker maintains its own copy of the Nyström feature matrix $\mathbf{Z}$. With $m = 500$ and $n = 90{,}000$,

each copy occupies approximately 360 MB, so 40 parallel workers require 14 GB—well within the 274 GB available on our workstation.